Second Year Engineering

23CSE3004-Operating System

Class - S.Y. (SEM-II)

Unit - I

INTRODUCTION TO OPERATING SYSTEM

AY 2025-2026 **SEM-II**

SYALLABUS

Introduction to Operating Systems: Evolution of Operating System, OS Definition, Operating System Overview – Objectives and Functions, Types of operating systems (Batch OS, Multi-programming OS, Time-sharing OS, Real-time OS), Functions, OS Structure(Monolithic, Microkernel, Layered, Client-Server), OS services (system calls, interrupt handling), System Programs, Kernel and user space, OS Generation and System Boot

Process Management: Process definition, the process concept – system programmer's view of processes – operating system's views of processes, States, Process Control Block (PCB), Process Creation and Termination (fork, exec, process hierarchy, exit, wait), Context switching, Types of schedulers (long-term, short-term, medium-term), Scheduling algorithms- Preemptive vs. Non-Preemptive Scheduling (FCFS, SJF, RR, Priority Scheduling)

Case study: CPU scheduling algorithms

Unit-I INTRODUCTION TO OPERATING SYSTEMS

What is an Operating System?

- Computer System = Hardware + Software
- **Hardware** □ only understand machine code i.e 0 and 1).
- **Software** □ Application Software + System Software(OS)
- An Operating System is a **system Software that acts as an interface between a user and the computer hardware.**
- Operating system is a collection or set of programs, which manages all the resources of the computer system.
- It basically manages all the resources of the computer.
- Example: Unix, Window, Linux, Ms-Dos, Mac, etc

What is an Operating System?

- Operating System is used as a communication channel between the Computer hardware and the user
- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system can also be defined as Resource allocator – manages and allocates Resources.
- Control program – controls the execution of user programs and operations of I/O devices.

Characteristics of OS

- Memory Management
- Resource Management (Device , File Management)
- Scheduling
- Security
- Interaction with different software's

Evolution of Operating System

First Generation Computers (1940-1950)

- The first computers used vacuum tubes(a sealed glass tube containing a near-vacuum which allows the free passage of electric current.) for circuitry and magnetic drums for memory.
- They were often enormous and taking up entire room.
- First generation computers relied on machine language. There were nothing such as OS.
- Was Using Serial Processing.
- They were very expensive to operate and in addition to using a great deal of electricity, generated a lot of heat, which was often the cause of malfunctions(defect or breakdown).
- The UNIVAC and ENIAC computers are examples of first-generation computing devices.

- **Advantages :**
- It was only electronic device
- No OS was present

- **Disadvantages :**
- Too bulky i.e large in size
- Vacuum tubes burn frequently
- They were producing heat
- Maintenance problems

Evolution of Operating System

Second Generation Computers (1950-1960)

- Transistors replaced vacuum tubes
- Used Batch System / Created base for multitasking.
- High-level programming languages were also being developed at this time, such as early versions of COBOL and FORTRAN.
- **Batch Processing:** Jobs were submitted in batches, and an operator loaded them into the computer sequentially.
- **Job Control Language (JCL):** Introduced to manage jobs.
- **No Interactive Processing:** Programs were executed without user interaction.
- **Examples:** IBM 701, IBM 7090.

Evolution of Operating System

Second Generation Computers (1950-196

IBM 1620

Advantages :

Size reduced considerably

Faster than previous generation

Disadvantages :

They over heated quickly

Maintenance problems

Evolution of Operating System

Third Generation Computers (1960-1979)

- **Multiprogramming:** Multiple programs could run simultaneously, improving CPU utilization.

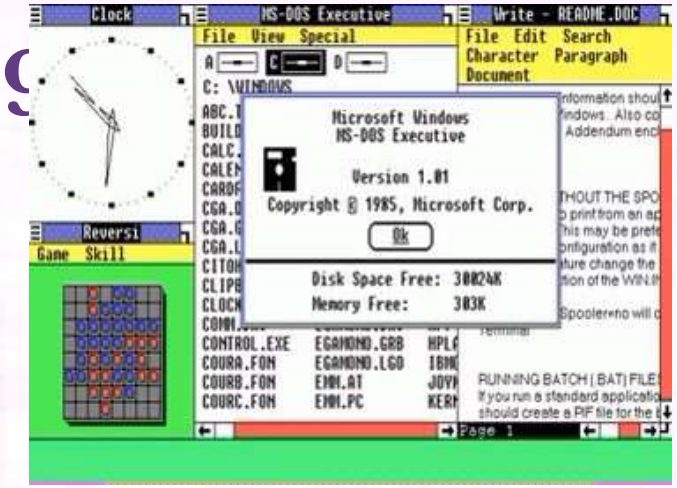
- **Time-Sharing:** Allowed multiple users to share computing resources interactively.

- **Introduction of System Software:** Operating systems like **UNIX** and **OS/360** emerged.

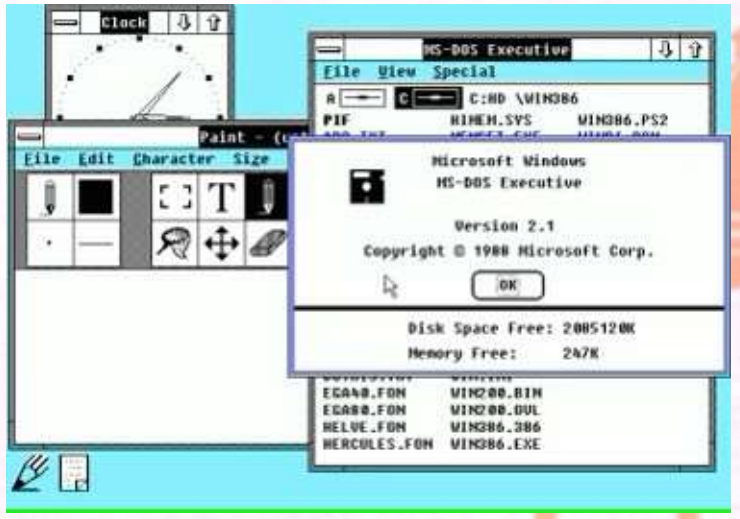
- **Features:** File systems, device drivers, and memory management.

Evolution of Operating System

Fourth Generation Computers (1980-199



- **Graphical User Interfaces (GUI):** Made systems more user-friendly (e.g., Windows and macOS).
- **Distributed Systems:** Introduced to connect multiple computers for resource sharing.
- **Networking Capabilities:** Supported protocols for communication over LAN
- **Examples:** MS-DOS, Windows 1.0, 2.0



Evolution of Operating System

Fifth Generation Computers (2000-Present)

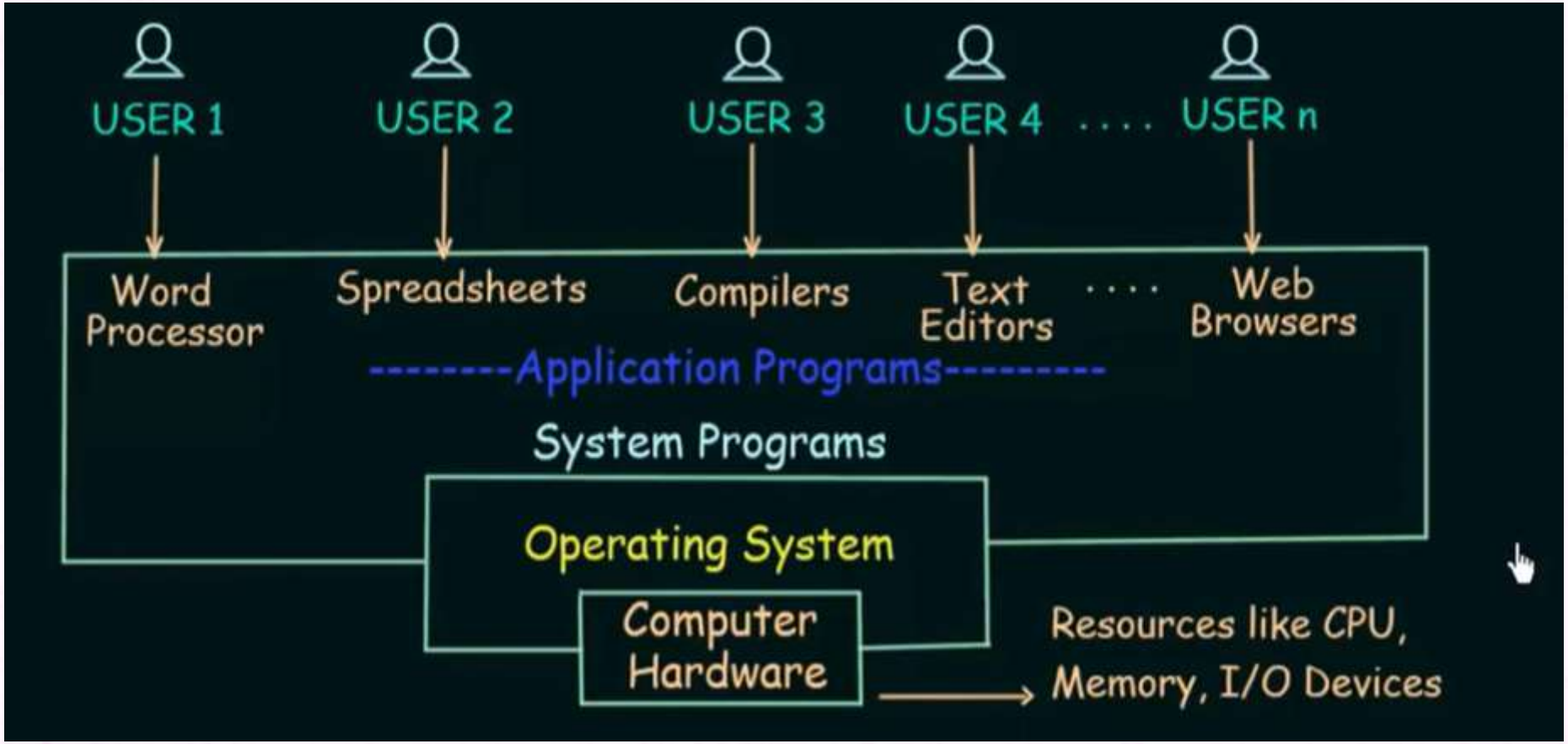
- **Mobile and Embedded Systems:** Operating systems optimized for mobile devices and IoT (e.g., Android, iOS).
- **Virtualization and Cloud Computing:** Virtual machines and cloud-based services became widespread.
- **Security and Scalability:** Focus on data security and scalable architectures.
- **Examples:** Linux, Windows 10, Android, iOS.

Evolution of Operating System

Summary

Generation	Key Features	Examples
1st	No OS, manual programming	ENIAC, UNIVAC
2nd	Batch systems, JCL	IBM 701, IBM 7090
3rd	Multiprogramming, time-sharing	UNIX, OS/360
4th	GUI, personal computing, networking	Windows 95, macOS
5th	Mobile systems, virtualization, cloud	Linux, Android, iOS

Overview of Operating Systems



Overview of Operating Systems

Divided into four components-

1] Hardware: Physical part of machine which provide basic computing resources.

Eg: CPU, Memory, I/O Devices etc.

2]Operating System:

- It controls & co-ordinates use of hardware among various application & user.
- It assign different jobs to the hardware part while running a program.

3]Application Program: It defines the ways in which the system resources are used to solve the computing problems of the users.

Overview of Operating Systems

-Application are readymade packages ready to use.
-They utilize the hardware as well as software resources and **fulfills user requirement.**
Eg: web browser, database system, video games etc.

4]User: Users are people, machines, other computers.

Characteristic of Operating System

- **Scheduling** allocate CPU to process
- **Memory Management**
- **Security** User access control to resources & monitors
- **Interaction with different software**
- **Device Management** I/O controlling
- **File Management** to store, organize, and manage files and directories

Advantages of Operating System

- **Source of computing** Input, processing and output
- **User-Friendly Interface-** GUI rather than command line interface.
- **Resource Sharing-** Sharing S/W & H/W recourses
- **No Coding Lines-** GUI eliminated the process of writing complex commands for doing small tasks.
- **Data Protection-** Username and Password, Biometric Verification, Authentication & recourse allocation
- **Multitasking-** ability to execute multiple tasks at a time

Disadvantages of Operating System

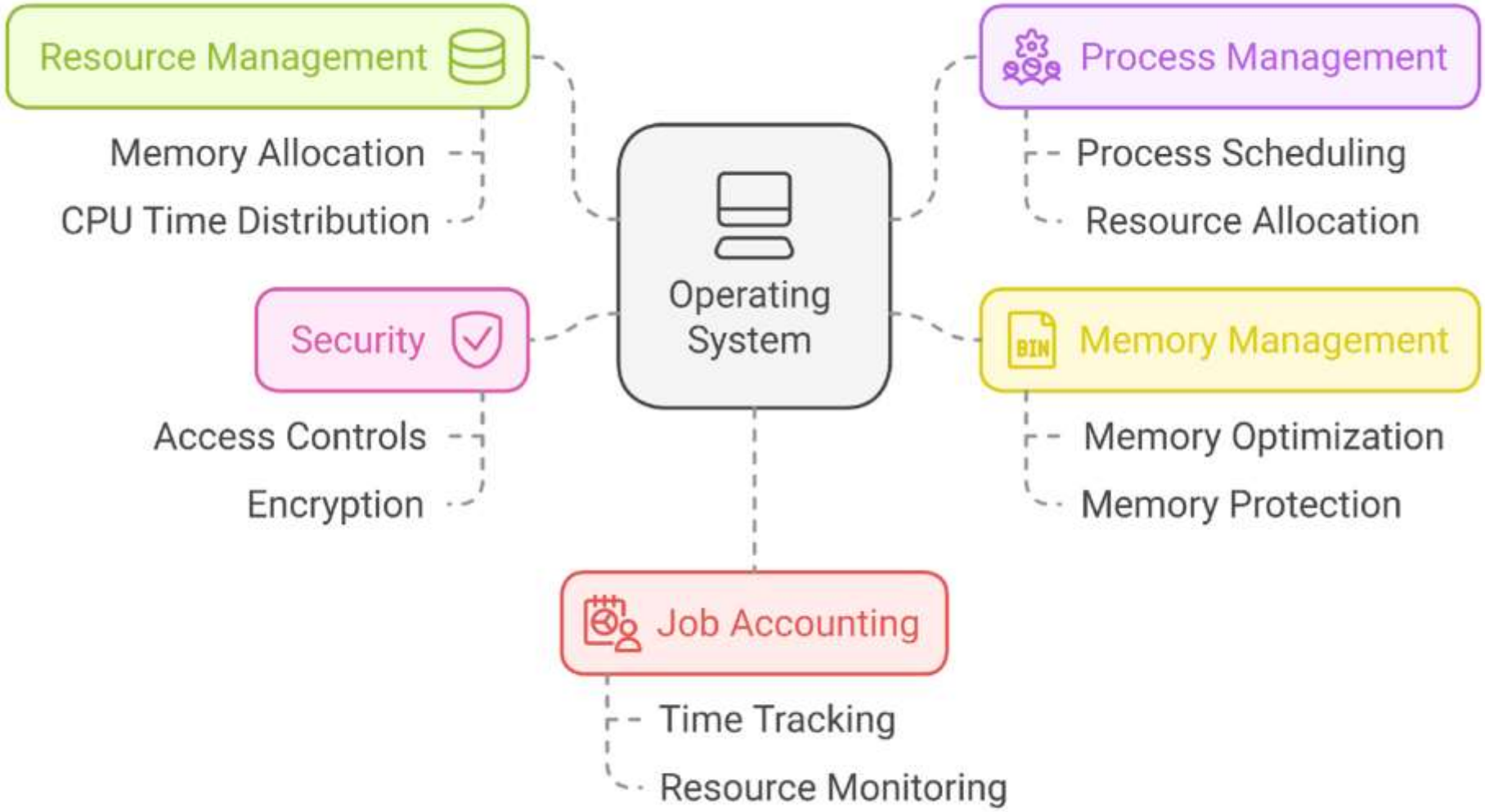
- **Expensive-**
- **System Failure-**
- **Highly Complex**

  language for programming
- **Virus Threats**

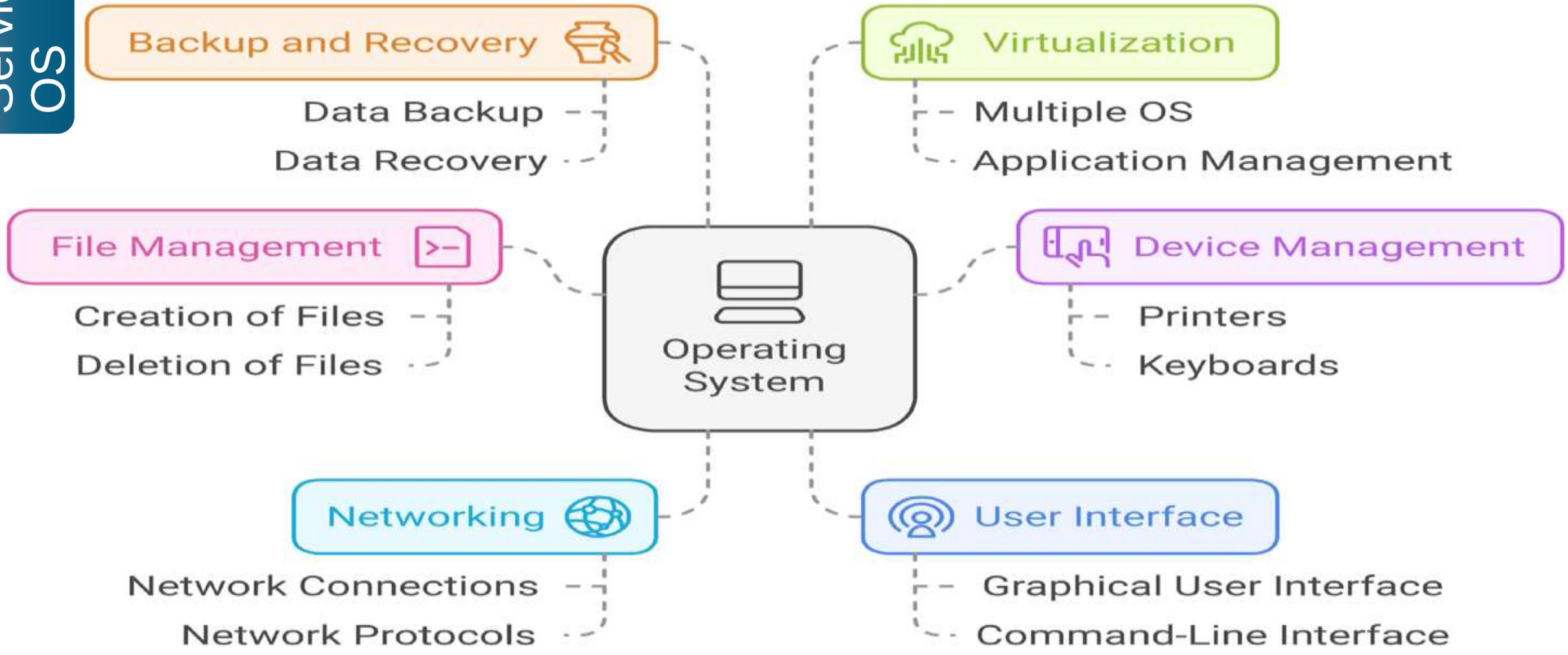
  halts the functioning and slows down.
- **Fragmentation**

- If an error occurred in your system, then there may be a chance that your data can be deleted therefore always have a backup of your data.
- Threats and viruses can attack our operating system at any time, making it challenging for the OS to keep the system protected from these dangers.
- For learning about new operating system can be a time-consuming and challenging, Specially for those who using particular Operating system for example switching from Windows OS to Linux is difficult.
- Keeping an operating system up-to-date requires regular maintenance, which can be time-consuming.
- Operating systems consume system resources, including CPU, memory, and storage, which can affect the performance of other applications.

Services of OS



Services of OS



Services of OS

- **Resource Management:** The operating system manages and allocates memory, CPU time, and other hardware resources among the various programs and processes running on the computer.
- **Process Management:** The operating system is responsible for starting, stopping, and managing processes and programs. It also controls the scheduling of processes and allocates resources to them.
- **Memory Management:** The operating system manages the computer's primary memory and provides mechanisms for optimizing memory usage.
- **Security:** The operating system provides a secure environment for the user, applications, and data by implementing security policies and mechanisms such as access controls and encryption.
- **Job Accounting:** It keeps track of time and resources used by various jobs or users.

Services of OS

- **File Management:** The operating system is responsible for organizing and managing the file system, including the creation, deletion, and manipulation of files and directories.
- **Device Management:** The operating system manages input/output devices such as printers, keyboards, mice, and displays. It provides the necessary drivers and interfaces to enable communication between the devices and the computer.
- **Networking:** The operating system provides networking capabilities such as establishing and managing network connections, handling network protocols, and sharing resources such as printers and files over a network.
- **User Interface**: The operating system provides a user interface that enables users to interact with the computer system. This can be a Graphical User Interface (GUI), a Command-Line Interface (CLI), or a combination of both.
- **Backup and Recovery:** The operating system provides mechanisms for backing up data and recovering it in case of system failures, errors, or disasters.
- **Virtualization:** The operating system provides virtualization capabilities that allow multiple operating systems or applications to run on a single physical machine. This can enable efficient use of resources and flexibility in managing workloads.

Goals of OS

Convenience (User-friendly): Makes the computer system easier and more intuitive for users to operate.

Efficiency: Maximizes resource utilization and performance, ensuring that the system runs
smoothly and quickly.

Portability: Allows the operating system to run on different hardware platforms with minimal modification.

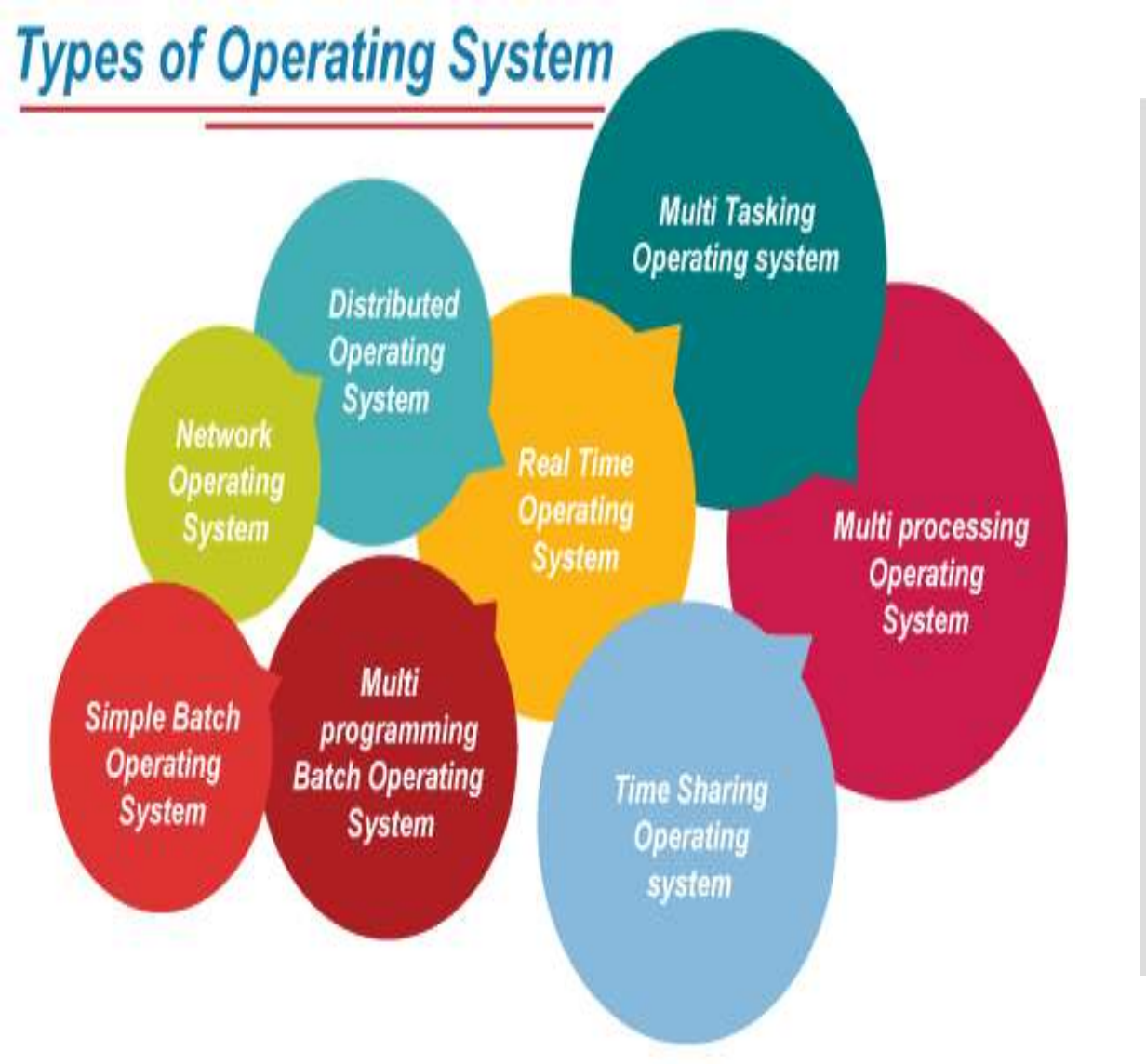
Reliability: Ensures the system operates consistently and correctly over time, with minimal
failures.

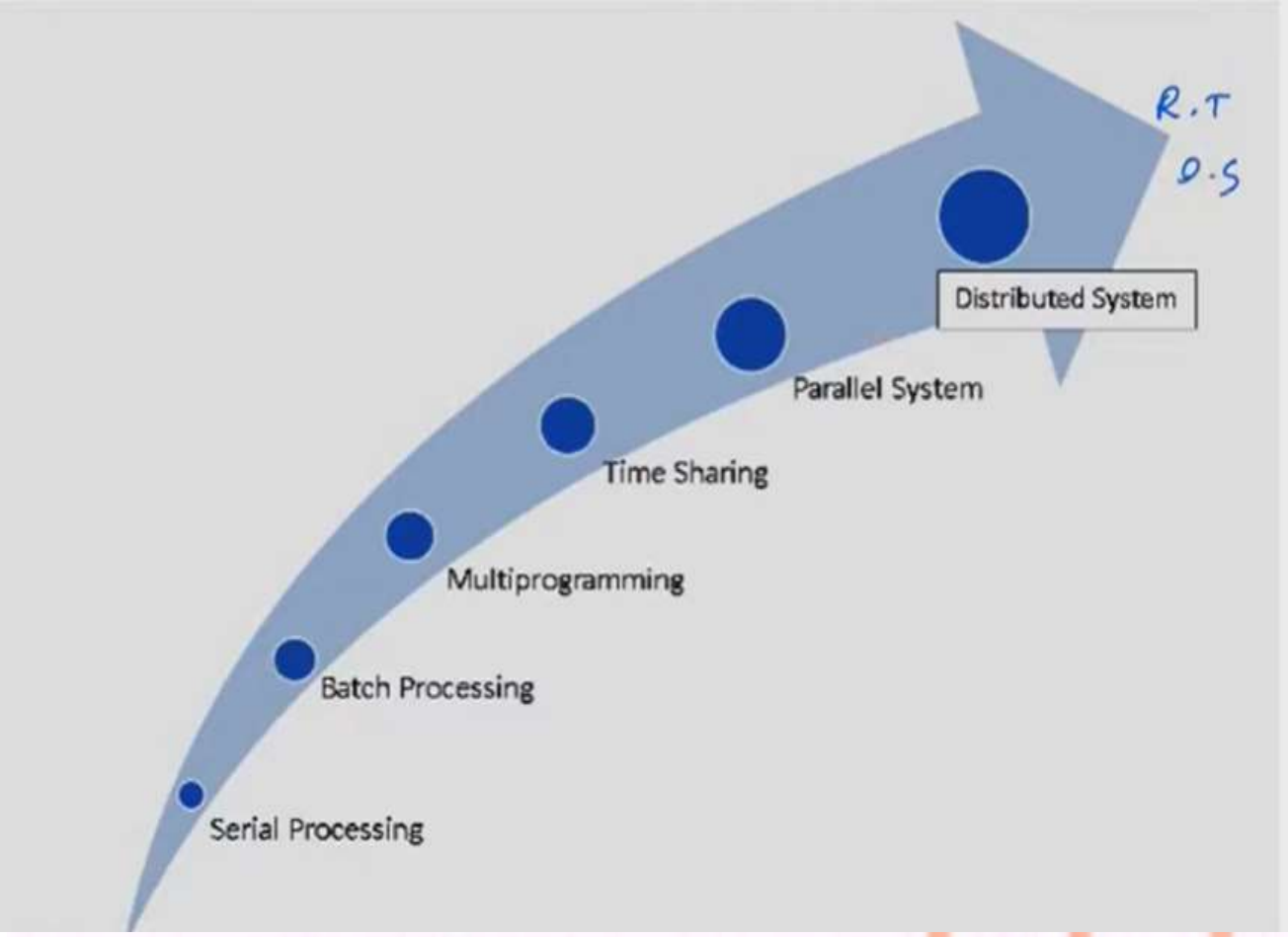
Scalability: Enables the system to handle increasing amounts of work or to be expanded to
accommodate growth.

Robustness: Maintains functionality and stability under various conditions,

Types of Operating Systems

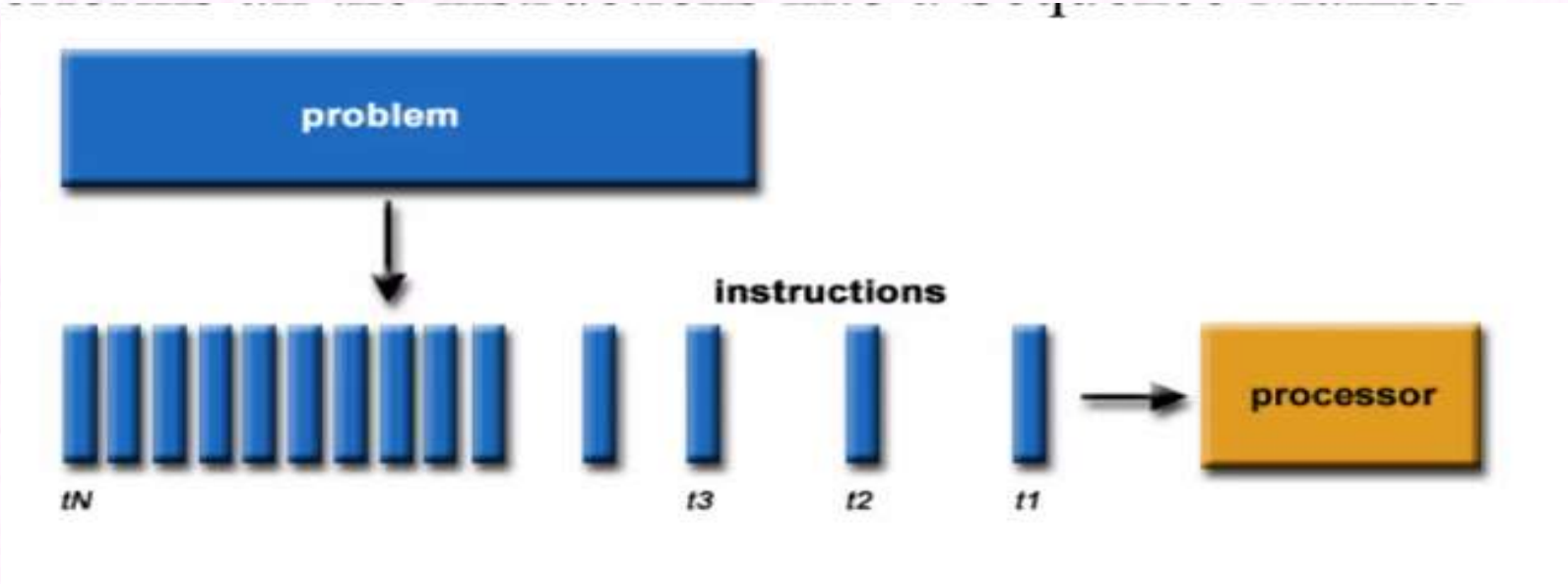
1. Batch Operating System
2. Time-sharing Operating System:
3. Distributed Operating System:
4. Network Operating System:
5. Real-time Operating System:
6. Multiprocessing Operating System:
7. Embedded Operating Systems:


Types of Operating System
Multi Tasking Operating system
Distributed Operating System
Network Operating System
Real Time Operating System
Multi processing Operating System
Simple Batch Operating System
Multi programming Batch Operating System
Time Sharing Operating system


R.T
O.S
Distributed System
Parallel System
Time Sharing
Multiprogramming
Batch Processing
Serial Processing

Serial Processing System

- No Operating system
- It works manually user/programmer
- User interacts with computer hardware directly
- Serial process system process one task at a time
- After completing one task, next task is being processed
- These computers were run from a console consisting of display. lights, toggle switches, some form of input device, and a printing device.



Drawback

Time Consuming

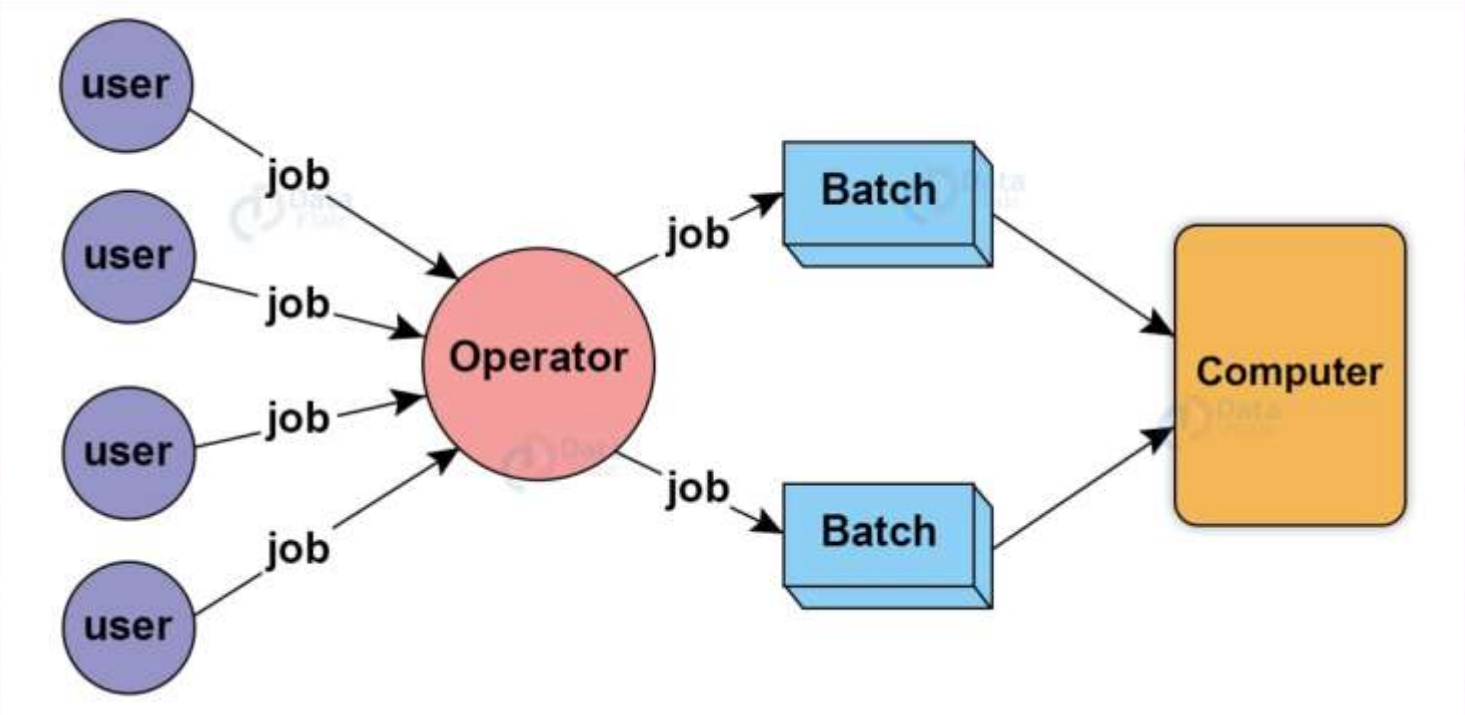
CPU is idle for long time

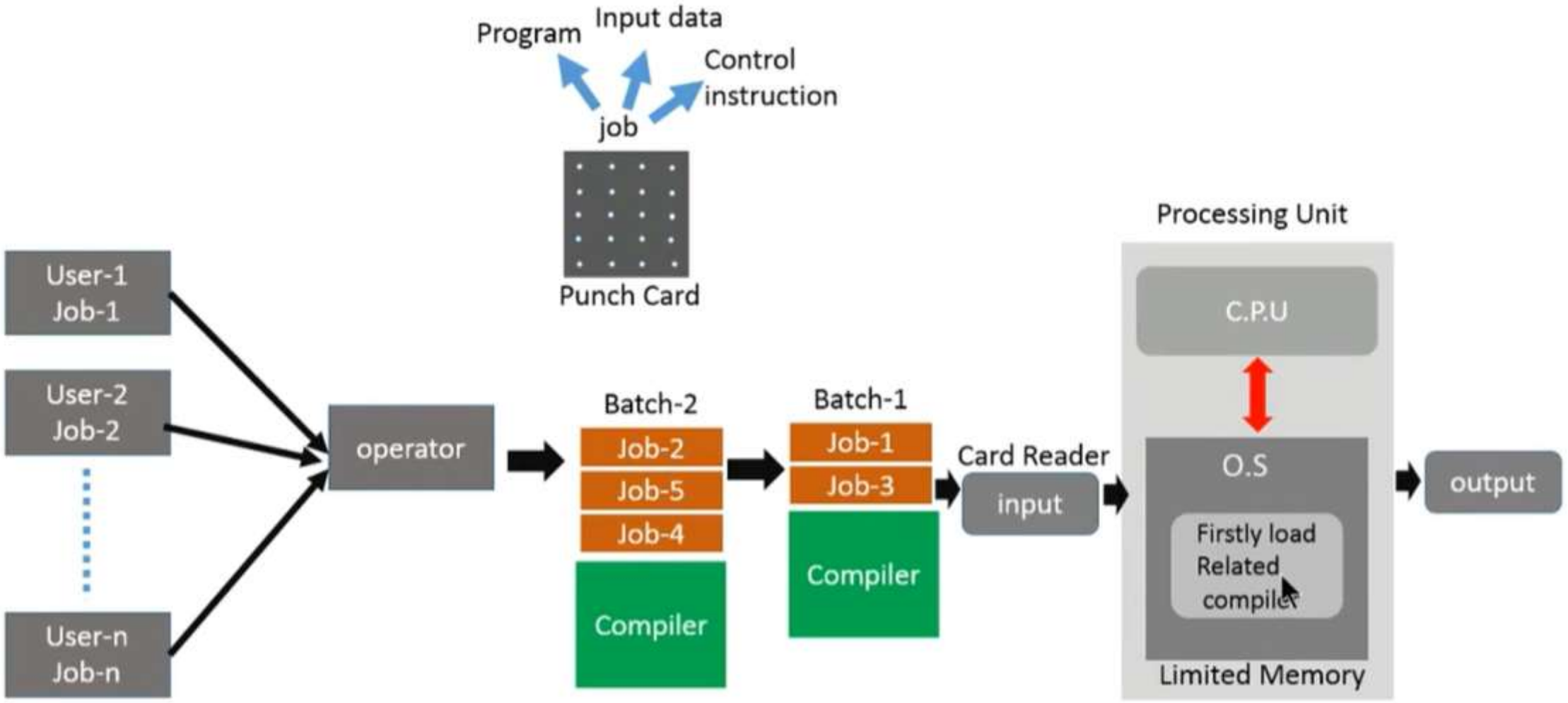
Lack of Interaction between user and his job/process

1. Batch Operating System

- A system that allows multiple users to use it, without direct communication between them.
- It achieves this by keeping all users in separate 'batches', meaning they can't interact with each other directly.
- This makes it ideal for applications where **users need to work on separate parts of a project, without getting in the way of each other.**
- **Examples of Batch Operating Systems:** Payroll Systems, Bank Statements, etc.

1. Batch Operating System




Program
Input data
Control instruction
job
Punch Card
User-1 Job-1
User-2 Job-2
User-n Job-n
operator
Batch-2
Job-2
Job-5
Job-4
Compiler
Batch-1
Job-1
Job-3
Compiler
Card Reader
input
Processing Unit
C.P.U
O.S
Firstly load Related compiler
Limited Memory
output

1. Batch Operating System

Advantages of Batch Operating System

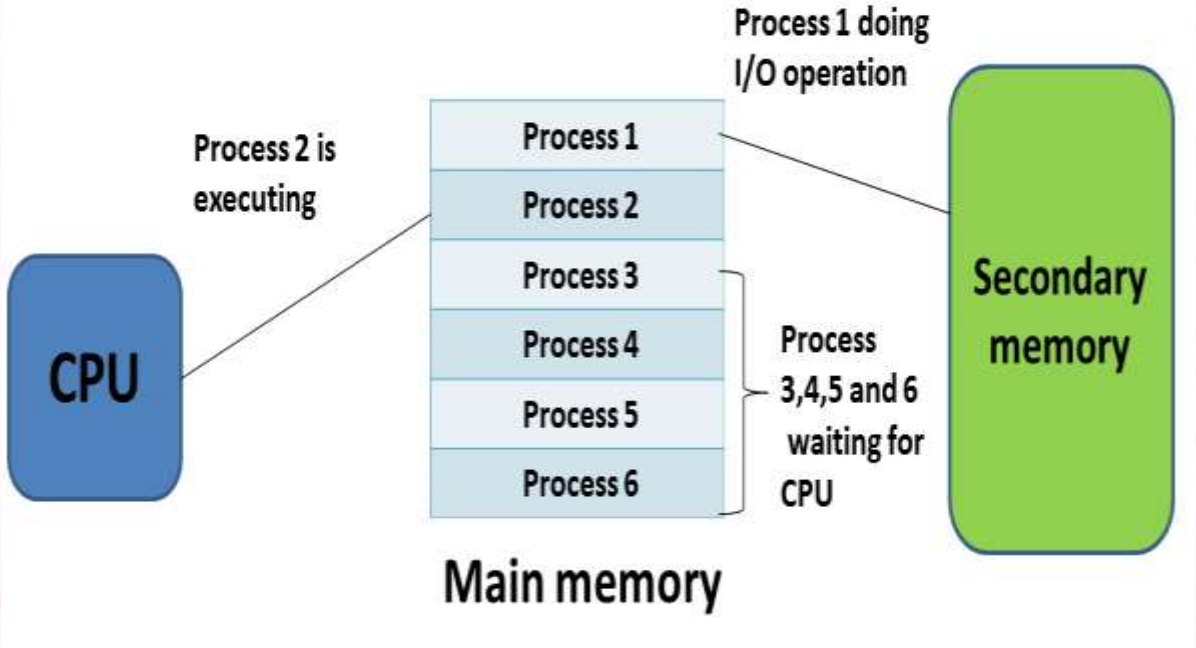
1. Multiple users can share the batch systems.
2. The idle time for the batch system is very less.
3. It is easy to manage large work repeatedly in batch systems.

Disadvantages of Batch Operating System

1. Lack of interaction between user and job
2. The operators should be well known with batch systems.
3. Batch systems are hard to debug.
4. The other jobs have to wait for an unknown time if any job fails.
5. It has a high turnaround time.
6. Difficulty to provide the desired priority
7. Failure of one job impacts the execution of other jobs

2. Multi-programming OS

The operating system which can run multiple processes on a single processor is called a multiprogramming operating system. There are different programs that want to get executed. So these programs are kept in the ready queue. And are assigned to the CPU one by one. If one process gets blocked then other processes from the ready queue are assigned to the CPU. The aim of this is optimal resource utilization and more CPU utilization. In the below figure, different processes are there in RAM(main memory). Some processes are waiting for the CPU, and process 2(which was previously executing) is now doing I/O operations. So CPU shifted to execute process 1.



Advantages of Multi-Programming Operating System

- CPU is better utilized and overall performance of the system improves.
- It helps in reducing the response time.

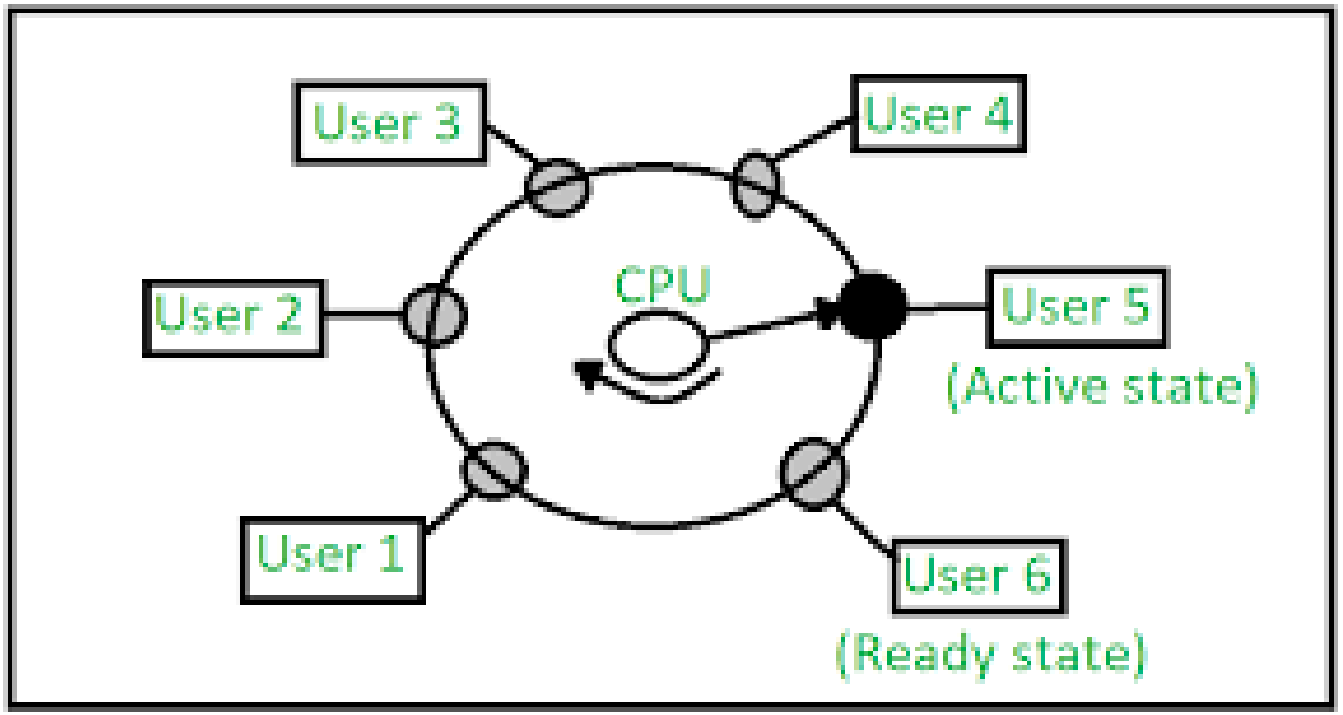
Disadvantages of Multi-Programming Operating System

- Prior knowledge of scheduling algorithms (An algorithm that decides which next process will get hold of the CPU) is required.
- If it has a large number of jobs, then long-term jobs will have to require a long wait.
- Memory management is needed in the operating system because all types of tasks are stored in the main memory.

3. Time-sharing Operating System:

- **Each task is given some time to execute** so that all the tasks work smoothly.
- **Each user gets the time of the CPU** as they use a single system.
- These systems are also known as Multitasking Systems.
- The task can be from a single user or different users also.
- The **time that each task gets to execute is called quantum.**
- **After this time interval is over OS switches over to the next task.**
- **Examples of Time-Sharing OS UNIX, Linux, IBM VM/CMS, Windows NT server, Windows 2000 server**

3. Time-sharing Operating System:



3. Time-sharing Operating System:

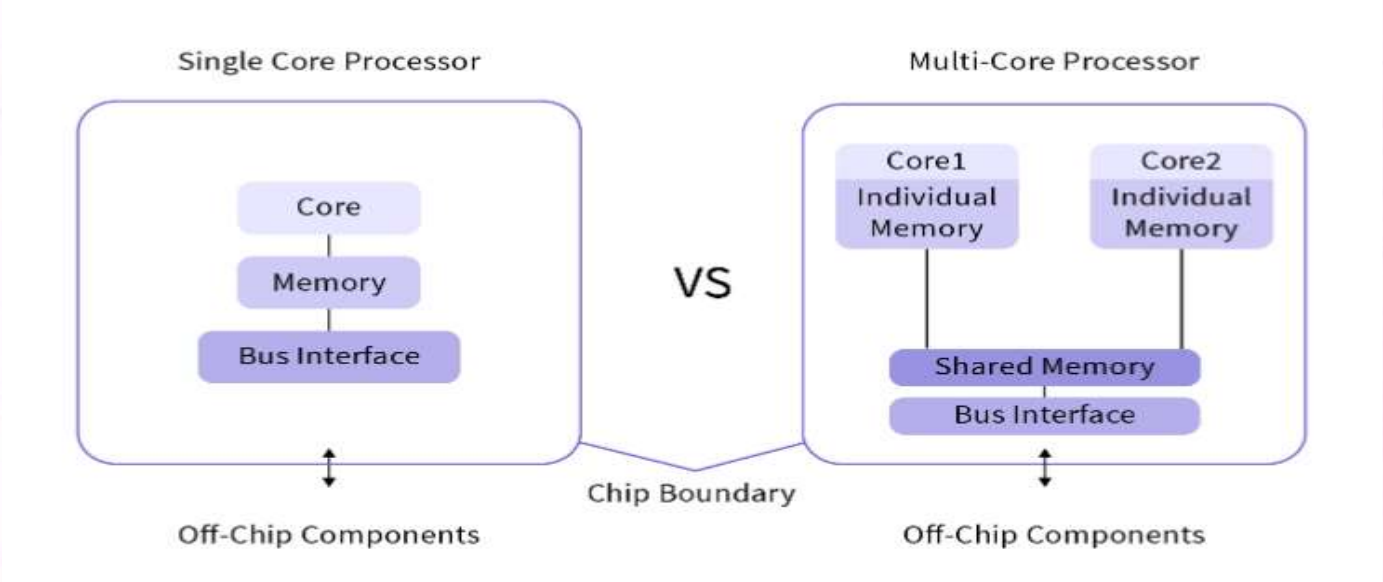
Advantages of Time-Sharing OS

1. Each task gets an equal opportunity.
2. CPU idle time can be reduced.
3. Resource Sharing between multiple users.
4. Time-sharing allows users to work concurrently. Its reduce waiting time.
5. Improved User Experience:

Disadvantages of Time-Sharing OS

1. It has difficulty with consistency.
2. A security and integrity problem with user programs and data.
3. Context switching overhead

4. Multiprocessing Operating System:



Single processor system

- **Throughput is less**
- **More cost** because require separate resources
- **No Fault tolerant**
- **Performance degrade with increased load**.

4. Multiprocessing Operating System:

- Distributes the workload among multiple processors to increase the efficiency and performance
- Is designed to manage and schedule tasks across multiple processors in a coordinated manner to maximize utilization and throughput
- linux, unix, windows MacOS
- High performance computing, scientific computing and server environment where the workload is large and parallel processing to achieve the optimal performance

4. Multiprocessing Operating System:

- It used to boost the performance of multiple CPUs within a single computer system.
- Multiple CPUs are linked together so that a job can be **divided and executed more quickly**.
- When a **job is completed, the results from all CPUs are compiled** to provide the final output.

Advantages

- Increased performance: throughput and response time
- Efficient utilization resources
- Improved scalability

Disadvantages

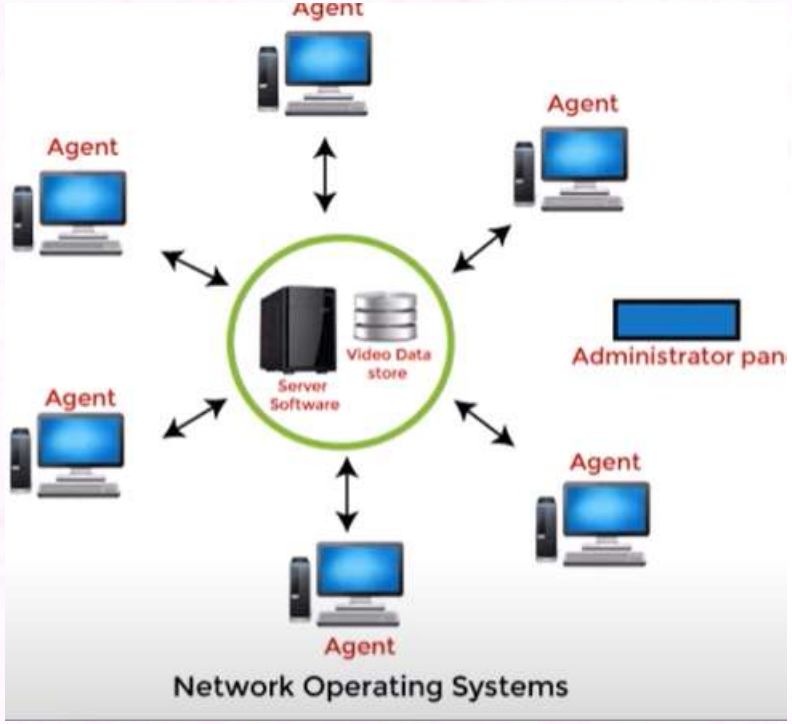
- Complexity-increase development and maintenance costs
- Resource contention : delay and decreased performance
- Synchronization overhead

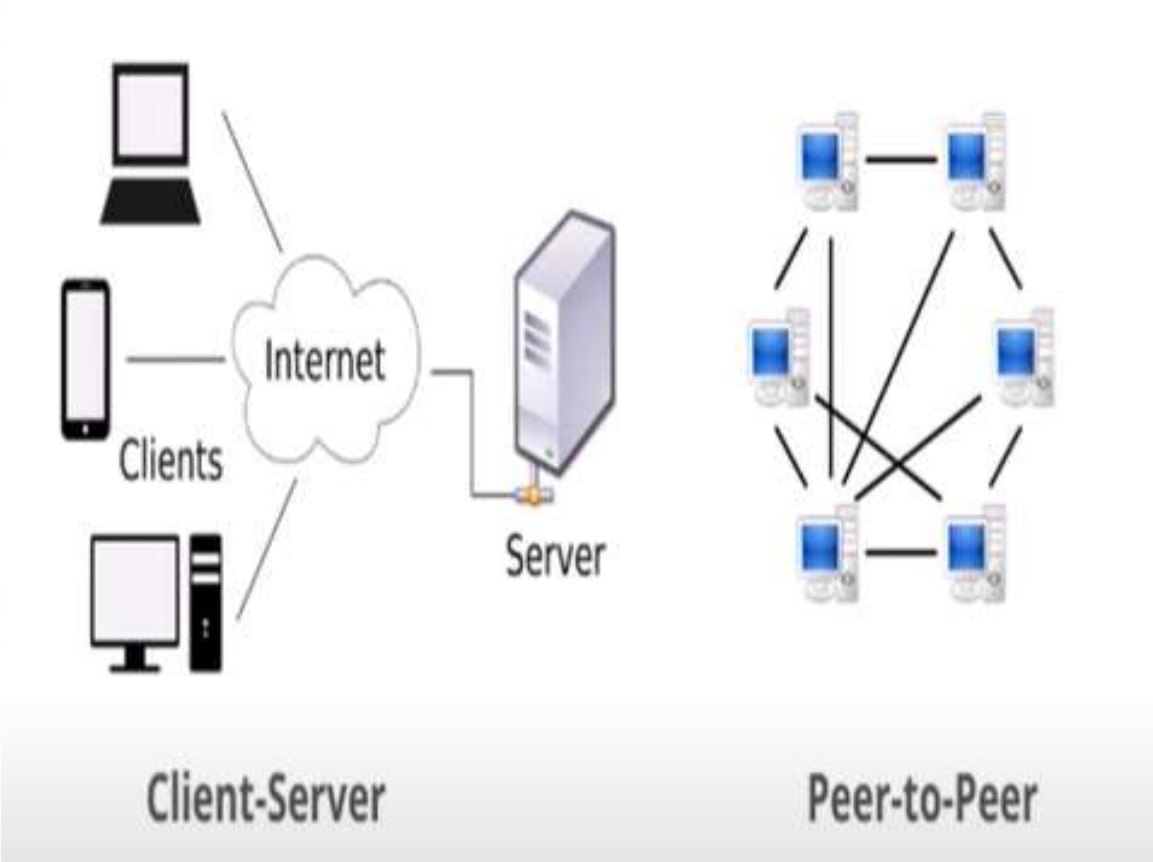
5. Network Operating System:

- An operating system that connects multiple computers and devices on a local area network to **allow sharing of files** and **resources by connection and communication over a network.**
- **Examples of Network Operating Systems** UNIX, Linux, Microsoft Windows Server 2008, Microsoft Windows Server 2003, Novell NetWare, Banyan's VINES, Artisoft's LANtastic, Mac OS X, and BSD.

5. Advantages of Network Operating System

1. Centralized management
2. Resource sharing
3. Security
4. Remote access





5. Disadvantages of Network Operating System

1. Complexity –
2. Cost -
3. Security –trained network administrator
4. Compatibility – different devices or software systems can’t work together as expected
5. Maintenance

5. Distributed Operating System:

- A **collection of autonomous computer systems** that are physically separated but are connected by a centralized computer network that is equipped with distributed system software.
- The autonomous computers will communicate among each system by sharing resources and files **and performing the tasks** assigned to them.
- **Distributed Operating Systems are of two types**: Client-Server Systems and Peer-to-Peer Systems.
- **Examples of distributed operating systems** Solaris, Micros, DYNIX, Ubuntu, Linux

3. Distributed Operating System:

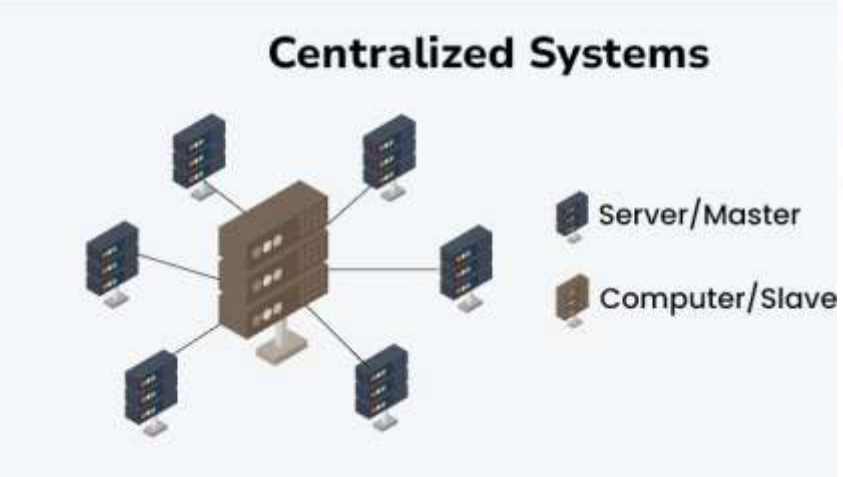
Advantages

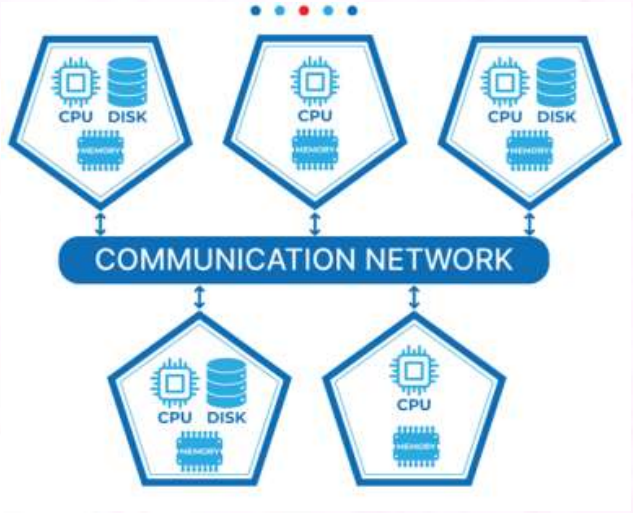
1. **Failure of one will not affect the other network communication**, as **all systems are independent** of each other.
2. **Load on host computer reduces.**
3. Since **resources are being shared, computation is highly fast** and durable.

Disadvantages

1. Failure of the main network will stop the entire communication.
2. To establish distributed systems the **language is used not well-defined yet.**
3. These types of systems **are not readily available as they are very expensive.**

3. Distributed Operating System:

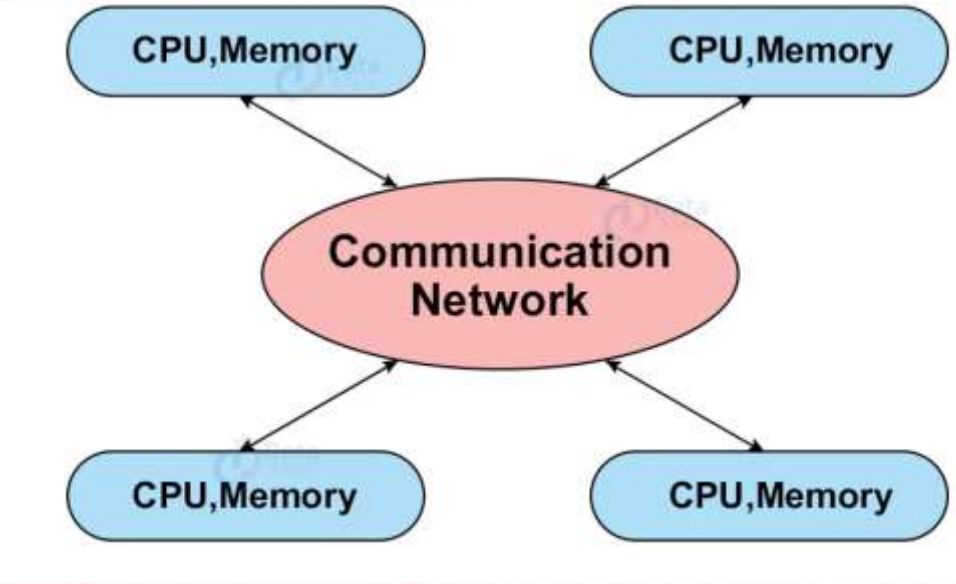


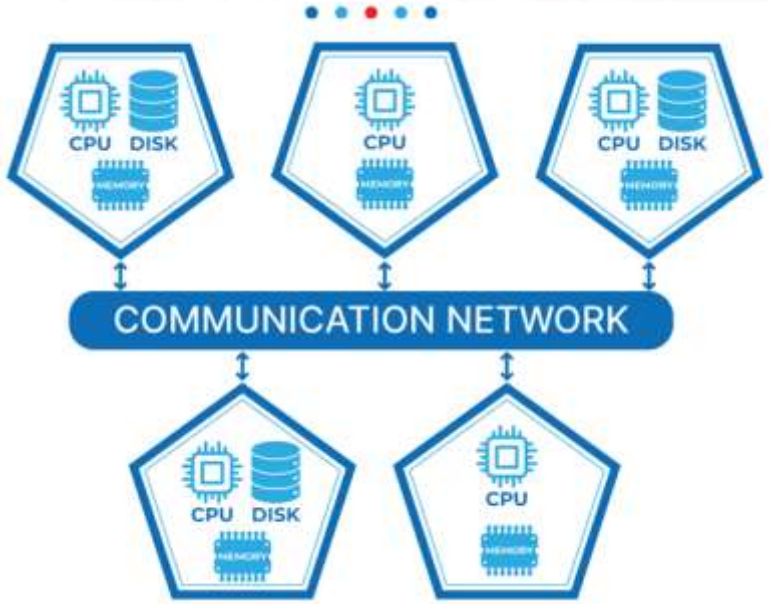


Key Characteristics of Centralized Systems

1. **Single Point of Control:**
2. **Scalability Issues:**
3. **Single Point of Failure**
4. **No Fault Tolerance**
5. **Coordination and Communication**
6. **Resource Sharing issues**
7. **Performance degrade with increased load.**

3. Distributed Operating System:





5. Embedded Operating Systems

1. Combination of Software & Hardware
2. Real-time Operating System
3. To Perform Specified Tasks
4. Reactive operation
5. Popular example are eCos, mbed OS, VxWorks, µC/OS-II, FreeRTOS

6. Embedded Operating Systems

- **Embedded Operating Systems** is a specialized operating system (OS) designed to perform a specific task for a device **that is not a computer.**
- The **main job of an embedded OS is to run the code that allows the device to do its job.**
- It makes the device's hardware accessible to software that is running on top of the OS.
- It typically **consists of a microcontroller or microprocessor, memory, input/output interfaces.**
- These **systems are programmed to carry out specific tasks and are often found in everyday devices such as appliances, automobiles, medical devices, and industrial machines.**

6. Advantages of Embedded Operating Systems

1. Cost:
2. Sizes:
3. Effectiveness:
4. Real-time operation:
5. Improved product quality.
6. low power operation
7. Portable due to the small size

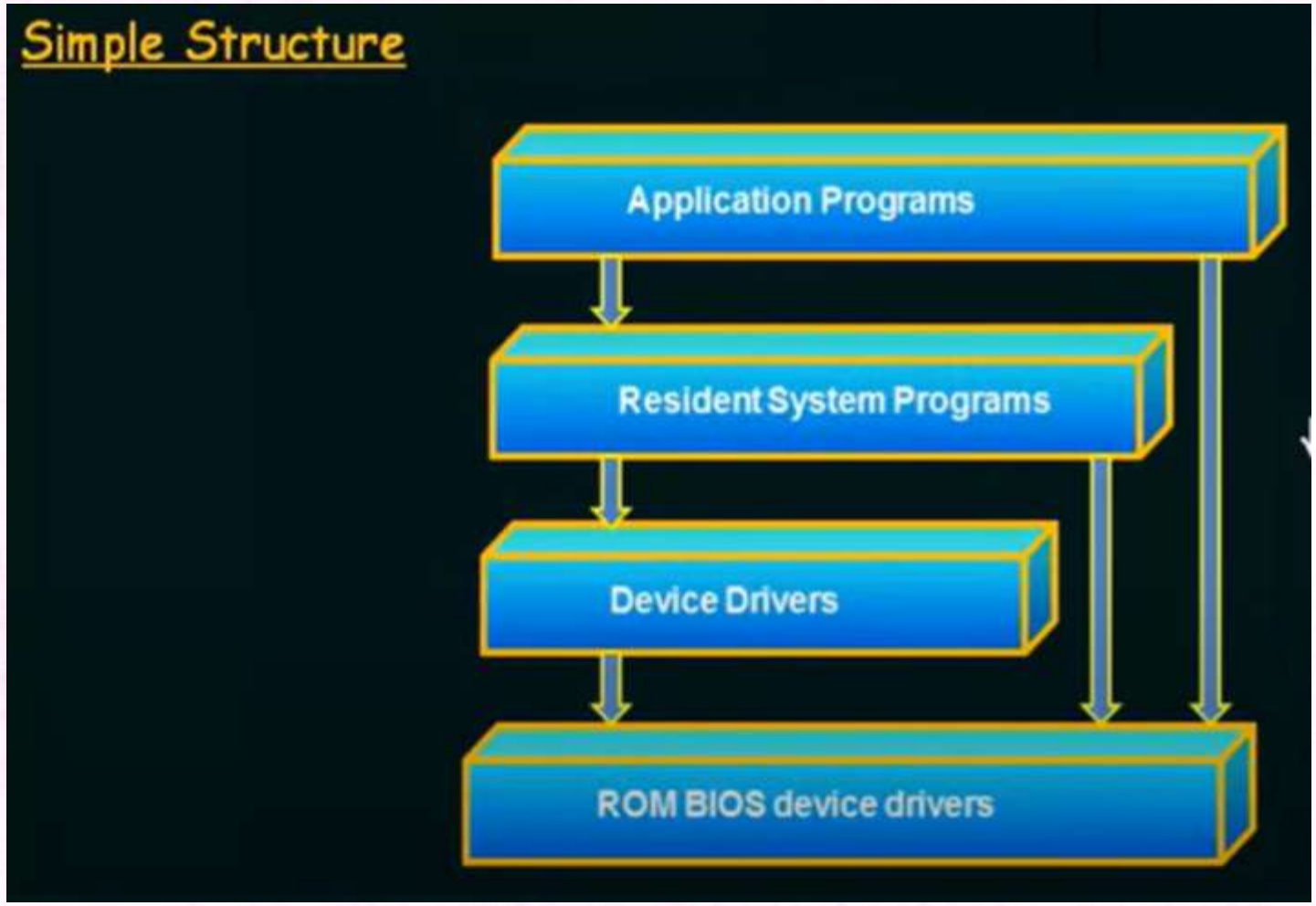
6. Disadvantages of Embedded Operating Systems

1. Limited Processing power
   - Resource constraints
   - Performance bottleneck
2. Limited flexibility and upgradability
   - fixed functionality
   - Hardware dependents
3. Development complexity
4. Cost of development
5. Scalability Issues:
6. Maintenance and Support:

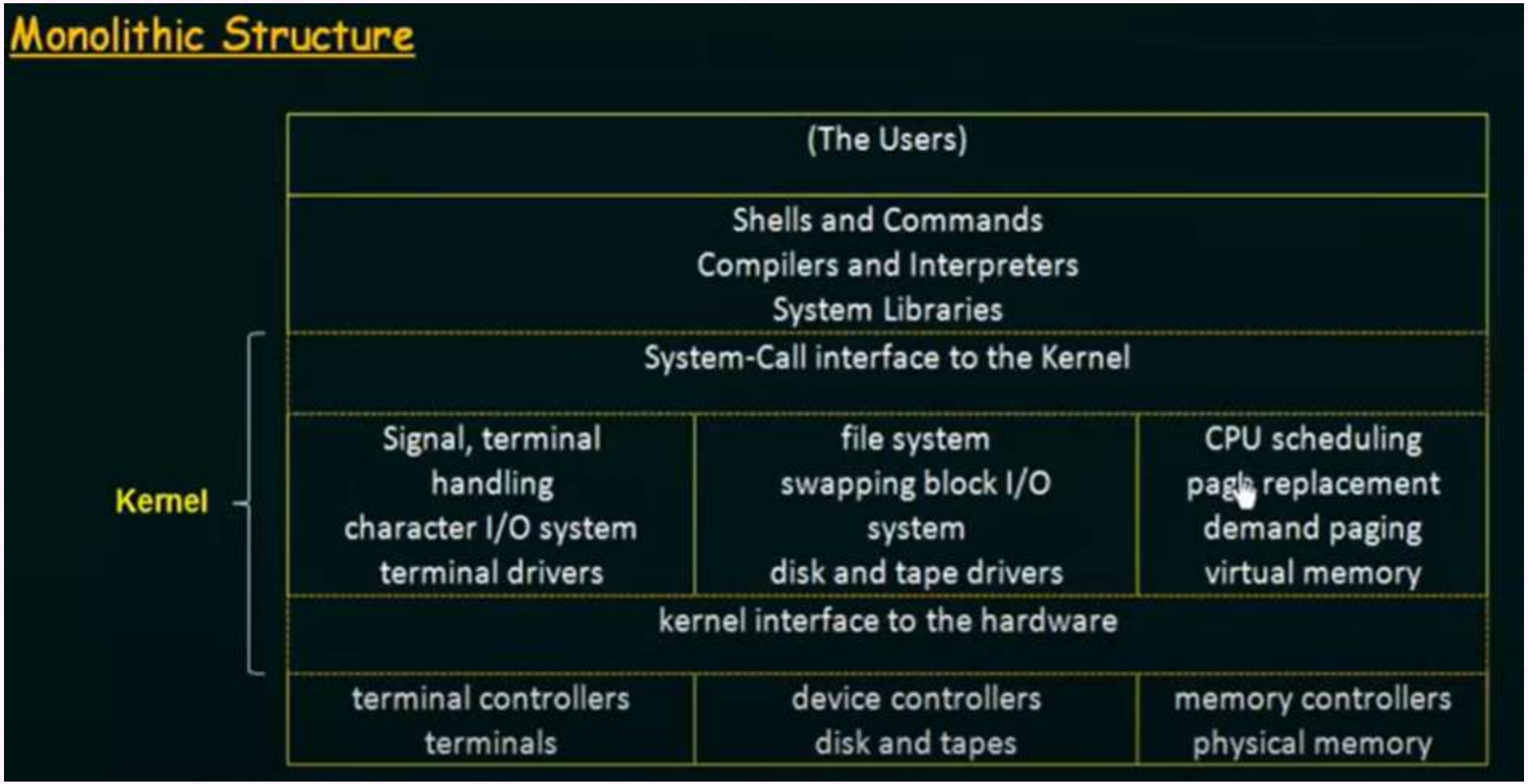
Computer-System Structure

1. Simple Structure
2. Monolithic Structure
3. Layered Structure
4. Microkernel Structure

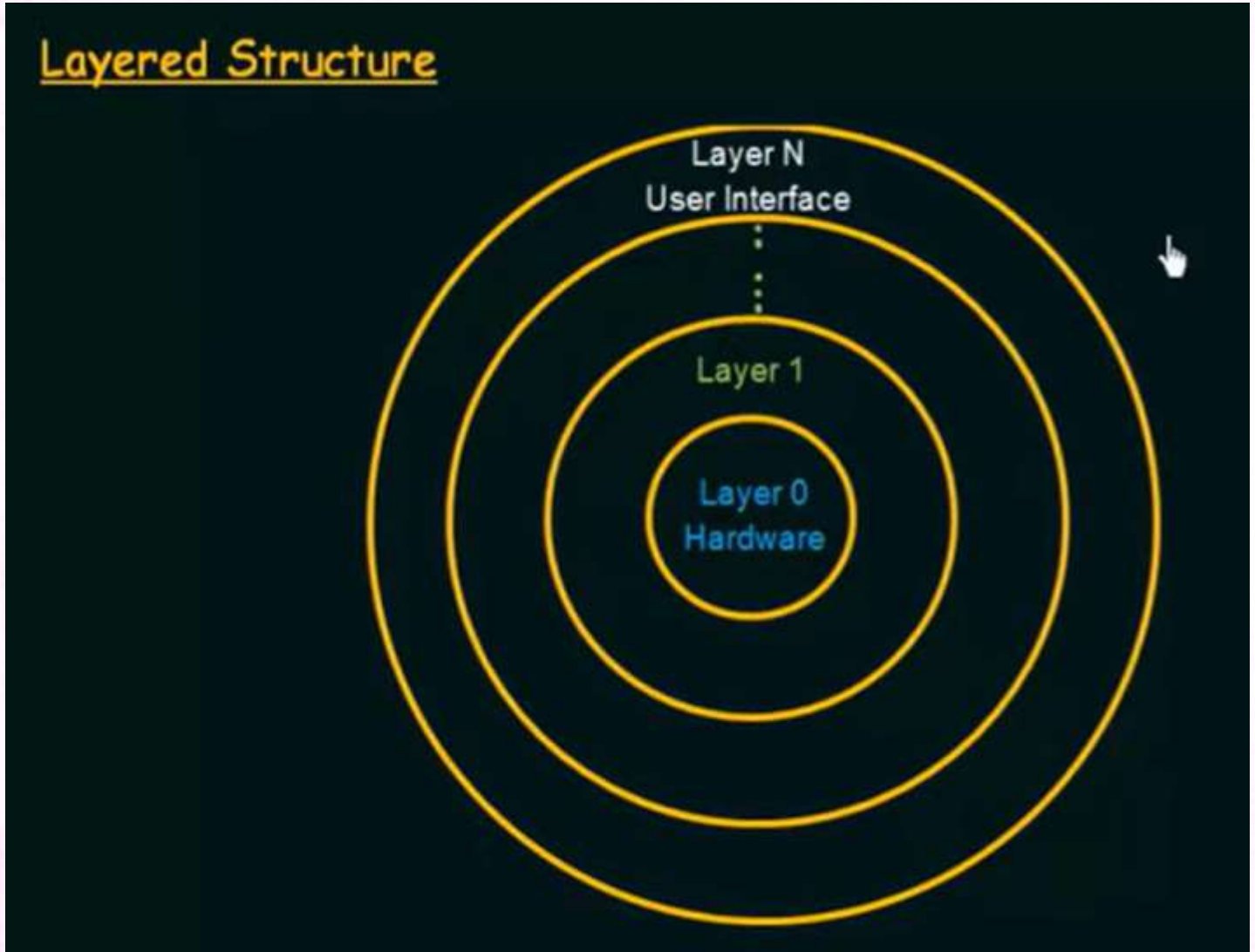
Computer-System Structure



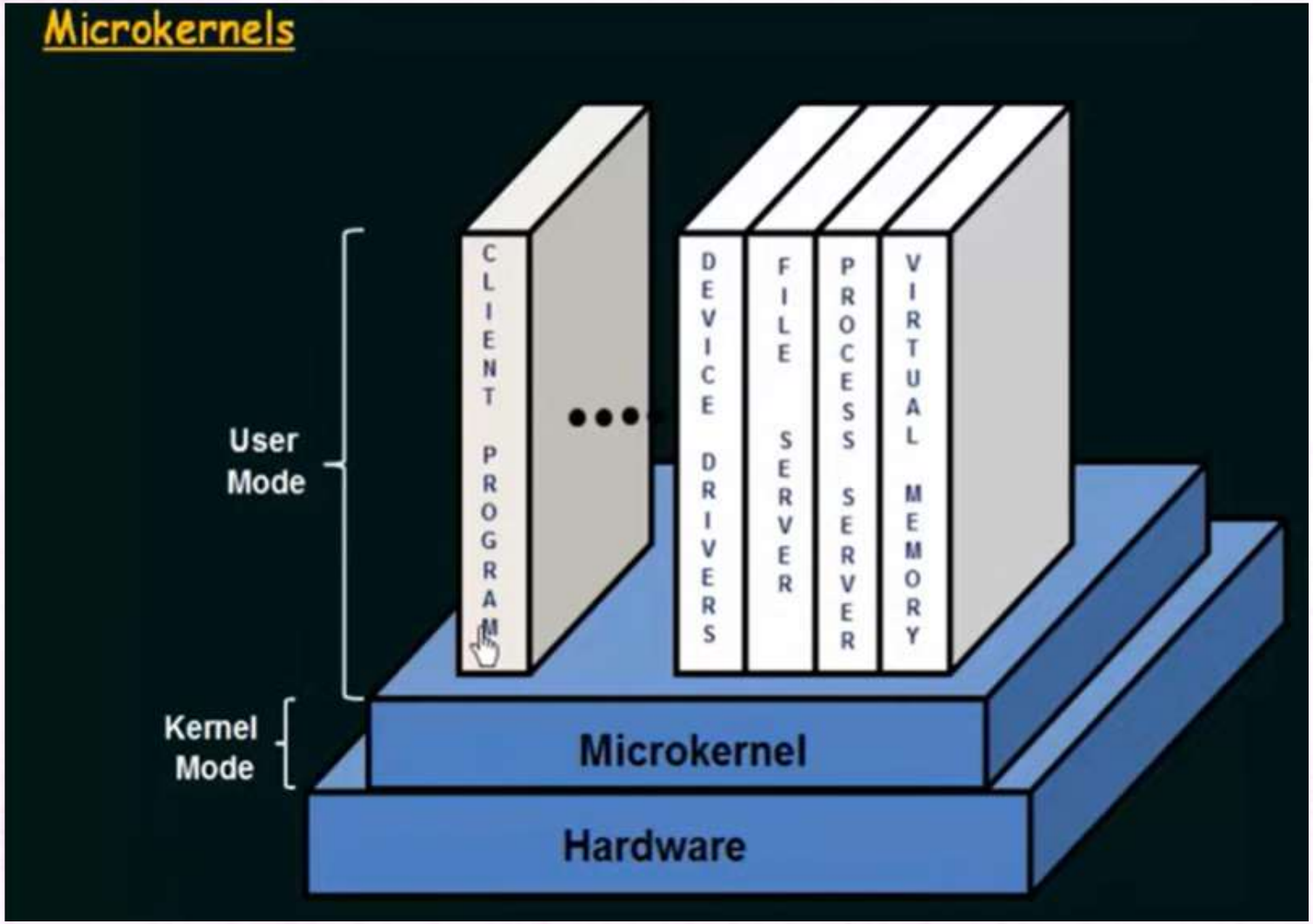
Computer-System Structure



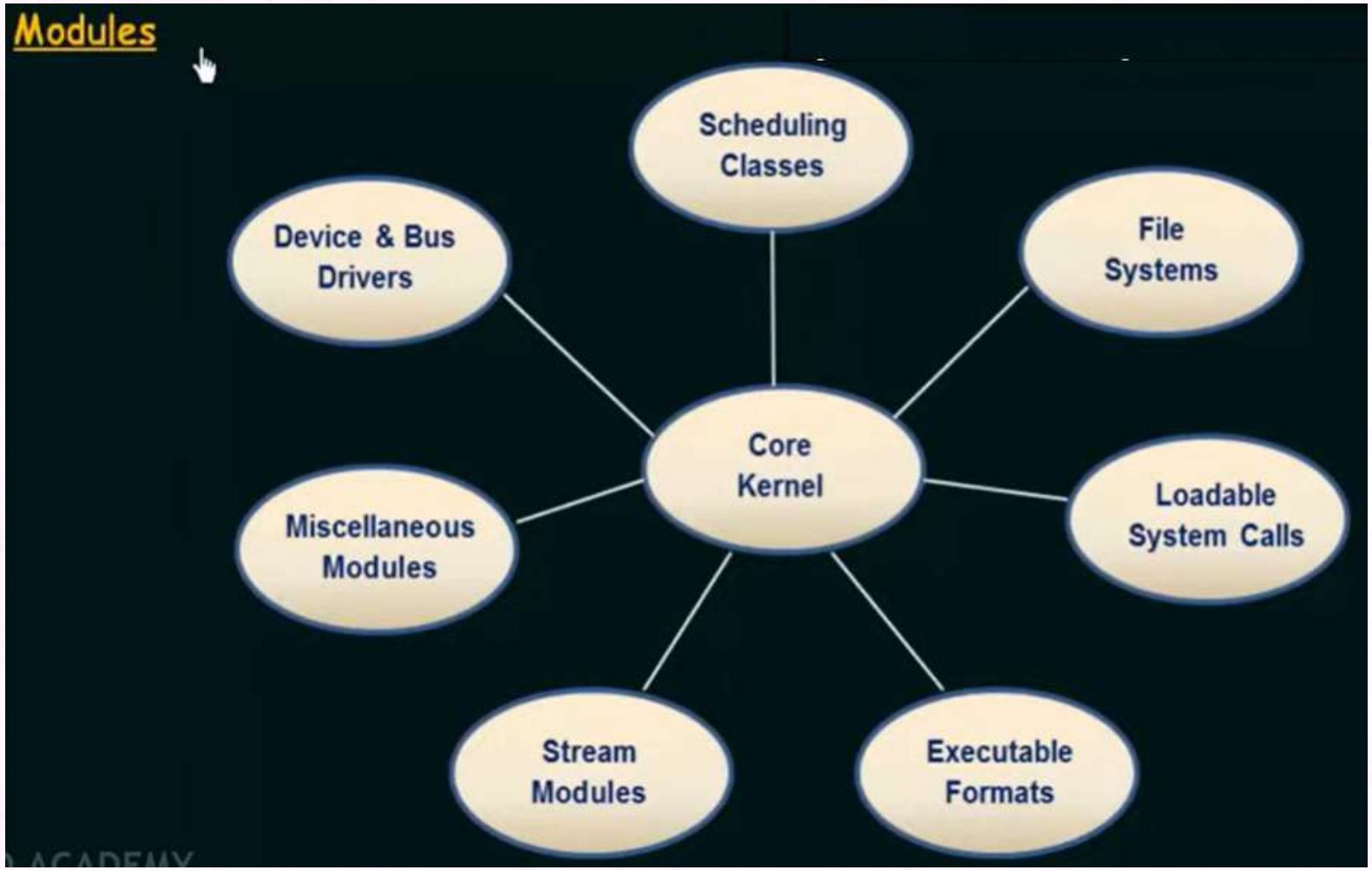
Computer-System Structure



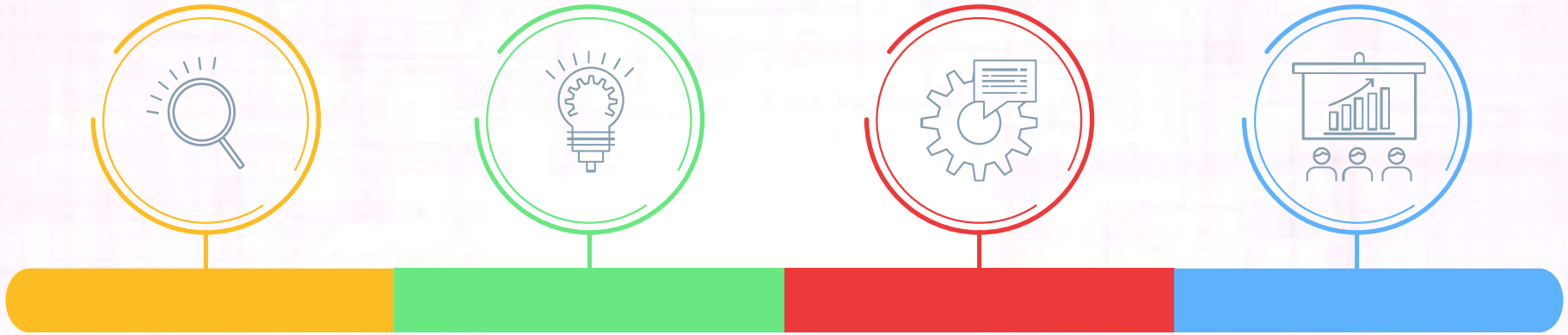
Computer-System Structure



Computer-System Structure



Computer-System Structure



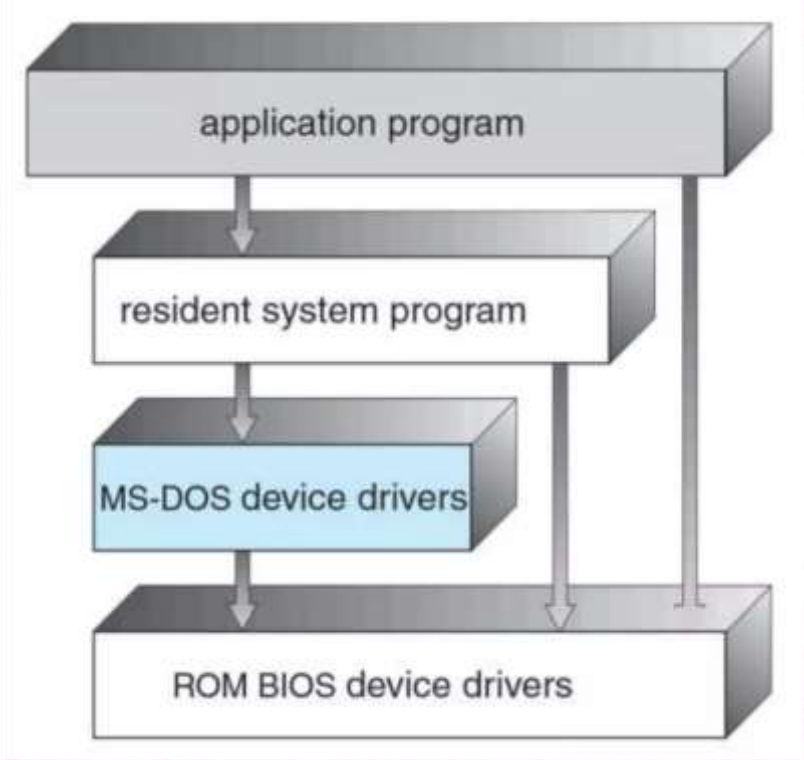
Simple Structure

Monolithic Structure

Layered Structure

Microkernel Structure

1. Simple Structure



- **Used very beginning** and **old Structure**
- **Not having well designed** structure
- **EX- MS-DOS**
- **At Bottom**□ **BIOS DD i.e HW**
- **DD access**□ **hw**
- **Resident system program**□ **DD & HW**
- **At Top Application Pragram**□ **Residencial system program** & Access HW.
- **Interface & levels of functionality** are **not separated.**
- **Used 8088**□ not support **dual mode** or not having **protection to hw.**

1. Simple Structure

- Not well design and structured.
- At bottom is ROM BIOS device driver is nothing but the Hardware layer.
- At top is application program is nothing but the user interface.
- Application program directly access hardware.
- Ex- MS-DOS

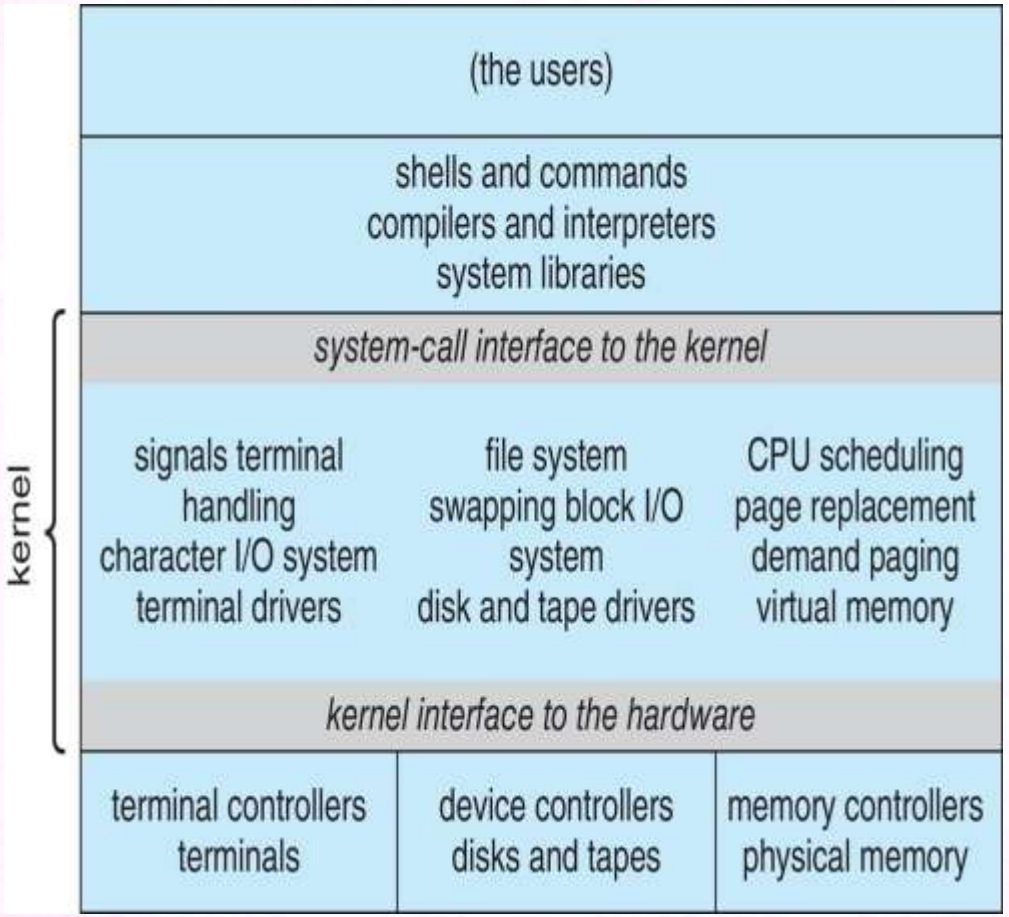
Advantages of Simple Structure:

1. simple to develop
2. better application performance

Disadvantages of Simple Structure:

1. Less Secure
2. If user application fails all os fails.
3. Not well design and structured.

2. Monolithic Structure



- **Followed by Unix OS**
- **1st □ User**
- **2nd □ Shell, system call, compiler, interpreter, system Library.**
- **3rd □ System call interface to kernel.**
- **5th □ above HW & below functionality.**
- **Everything packed in one level** hence called **monolithic structure.**
- **5th □ HW**

2. Monolithic Structure

- This operating system works in the kernel space in the monolithic system.
- The monolithic kernel is quite fast as the services such as memory management, file management, process scheduling etc.
- A set of primitives or system calls implement all operating system services such as process management, concurrency, and memory management.
- Example Linux, BSDs, Solaris, DOS, OpenVMS

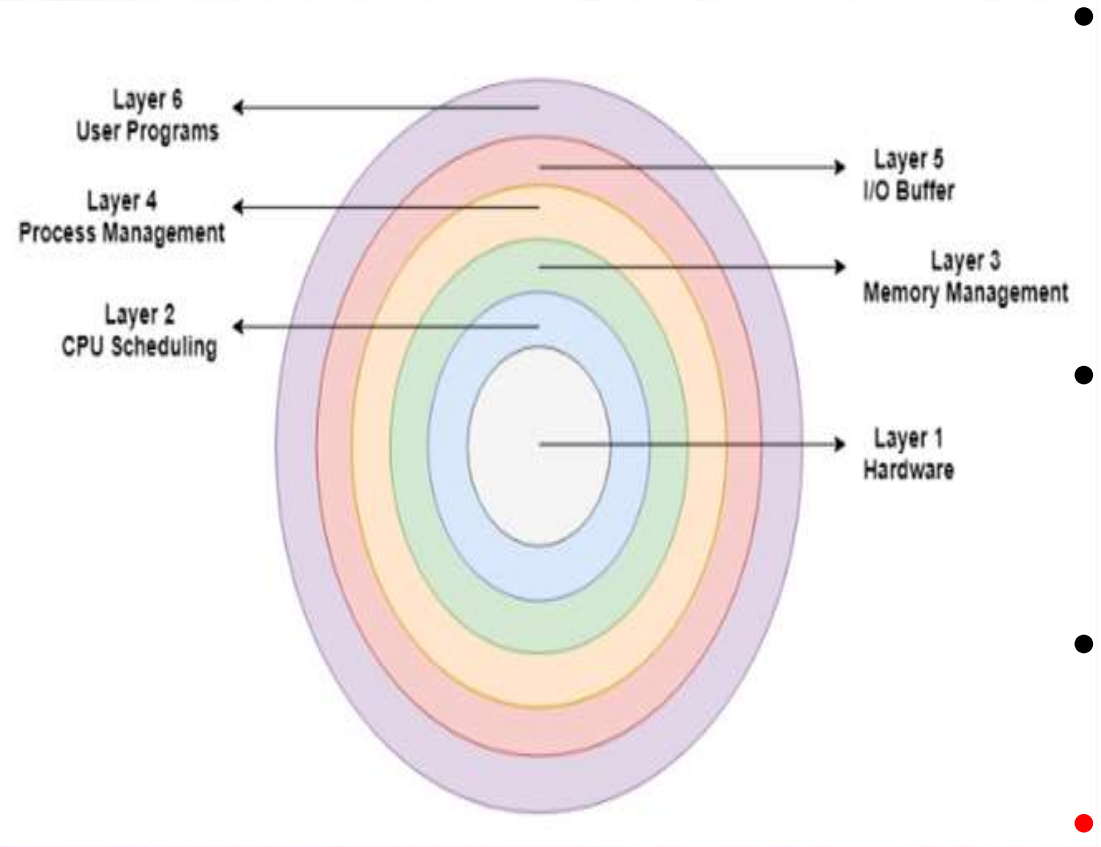
Advantages of Monolithic Kernel

- **Faster execution due to direct access** to all the services.
- **Simple and easy to implement structure**.
- **Fewer lines of codes need** to be **written** for a monolithic kernel.
- In memory massively quicker
- Less code to write

Disadvantages of Monolithic Kernel

- If one component fails, the entire system crashes
- Slower development speed –
- Debugging is complex and difficult
- It’s difficult to add new functionalities

3. Layered Structure



- **Divide into Nor of layers.** i.e **layer 0** □ **HW**…**layer N** □ **user interface.**
- **Broken down functionality** into separate layer.
- **Easy to implements** and **design.**
- **Ex- Windows NT OS**

Layered Structure

- Layered Structure is a type of system structure in which the different services of the operating system are split into various layers.
- Where each layer has a specific well-defined task to perform.
- It was created to improve the pre-existing structures like the Monolithic structure (UNIX) and the Simple structure (MS-DOS).
- Example – Windows NT operating system

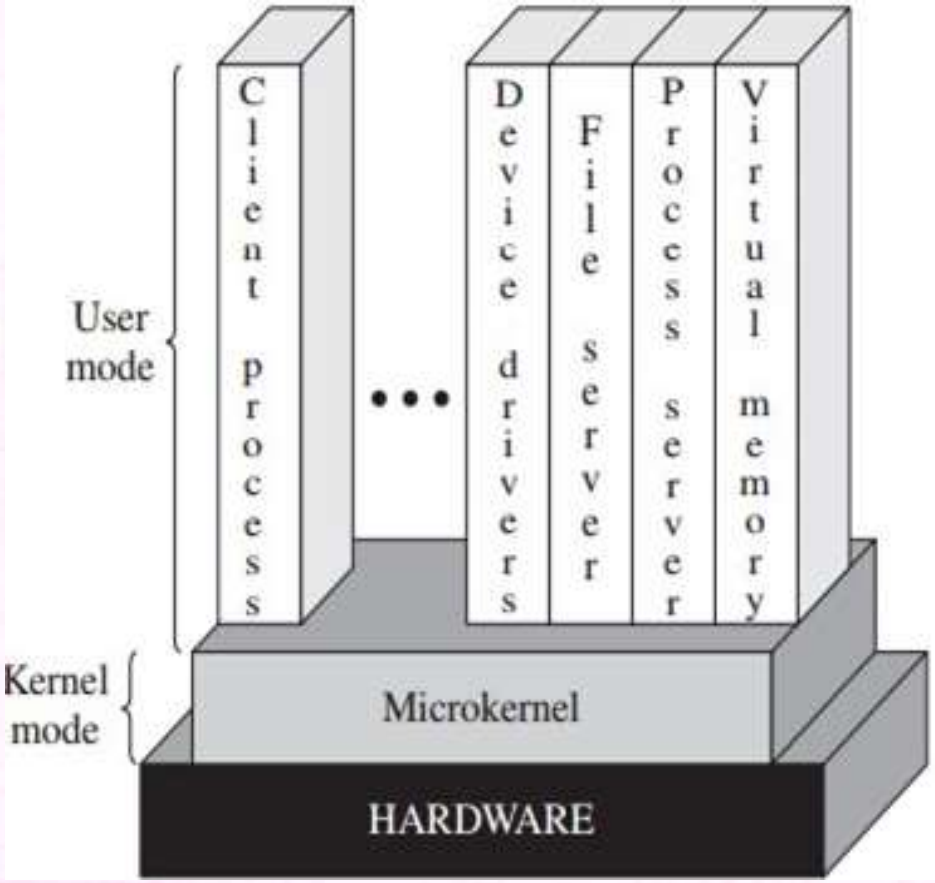
Advantages of Layered Structure

- Modularity:
- Easy debugging:
- Easy update:
- No direct access to hardware:
- Well design and define structure

Disadvantages of Layered Structure

- Complex and careful implementation:
- Slower in execution:
- Communication:(between adjacent layer)

4. Microkernel Structure (Ex- Mac OS X)



- In **monolithic kernel** all **functionalities packed** into kernel make structure big.
- **Micro □ Small structure**
- **Two separate mode**
- **User mode**□ essential function.
- **Kernel mode** □ Non-essential function.(**Core functionalities**)
- **Main function:- provide communication** bet client and services.□ **Massage passing**

4. Microkernel Structure

- In microkernel, user services and kernel services are kept in separate address space.
- OS is complex to design.
- Microkernel are smaller in size.
- Easier to add new functionalities.
- Failure of one component does not effect the working of micro kernel.
- Example : Mac OS X.

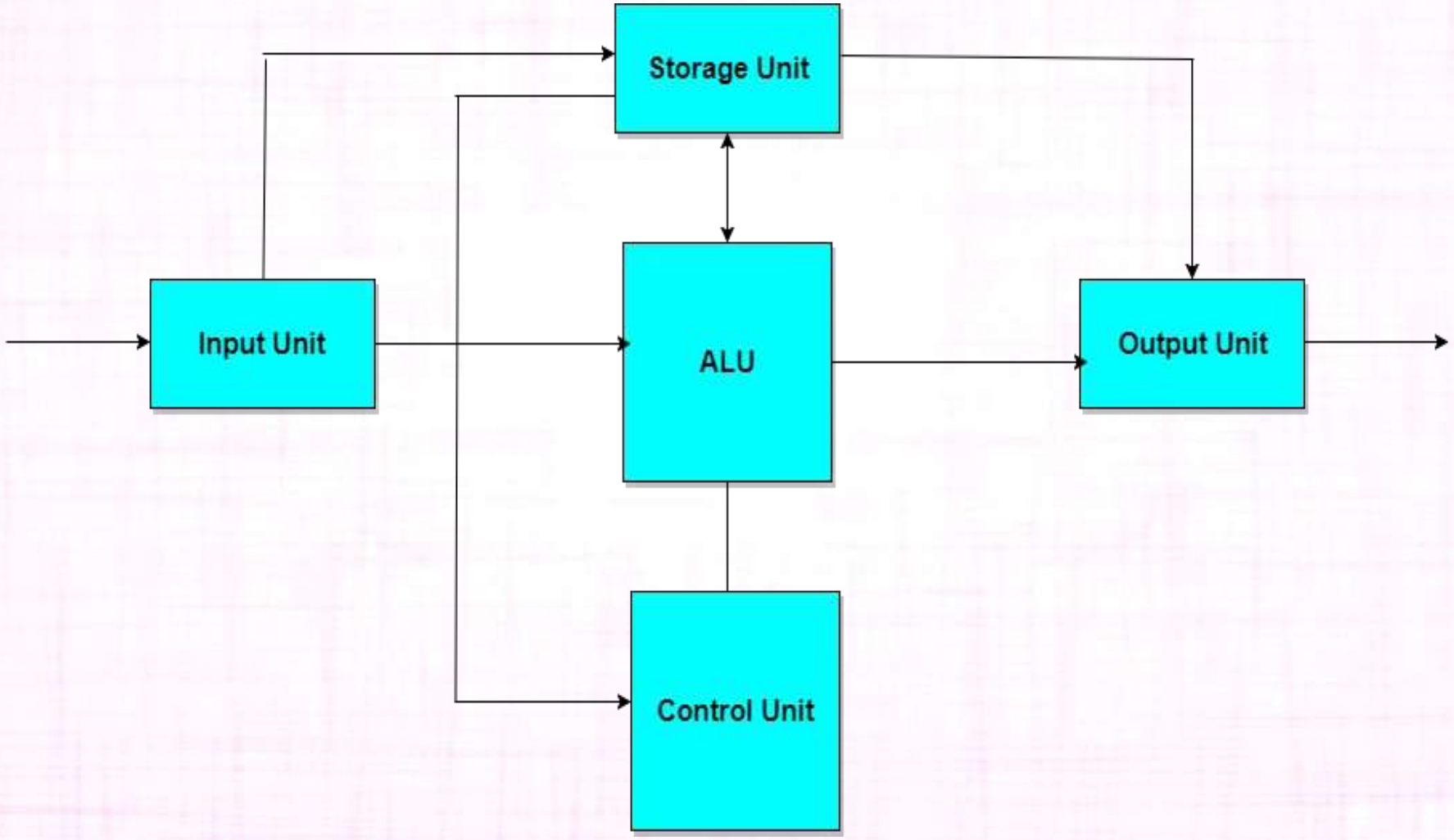
Advantages of Microkernel Structure

- It **allows the operating system** to be **portable between platforms**.
- Micro-Kernels are **smaller, they can be successfully tested.**
- **Failure of one component** does **not effect the working** of micro kernel.
- It is simple to maintain.
- **Easier to add** new functionalities.

Disadvantages of Microkernel Structure

- Complex to design Modular Structure
- **If Kernel fails** then then all system collapse
- Microkernel **system is costly.**
- **Low performance Massage passing**

Computer-System Architecture



Computer-System Architecture

- A computer system is basically a **machine that simplifies complicated tasks.**
- It should maximize performance and reduce costs as well as power consumption.
- The different components in the Computer System Architecture are

1. Input Unit,
2. Output Unit,
3. Storage Unit,
4. Arithmetic Logic Unit,
5. Control Unit etc.

Input Unit,

- The **input unit provides data** to the computer system from the outside.
- So, basically it **links the external environment with the computer.**
- It takes **data from the input devices, converts it into machine language** and then **loads it into the computer system.**
- **Keyboard, mouse** etc. are the most commonly used input devices.

Output Unit

- The **output unit provides the results** of computer process to the users i.e it links the computer with the external environment.
- Most of the output data is the **form of audio or video.**
- The different output devices are **monitors, printers, speakers, headphones** etc.

Storage Unit

- It is traditionally divided into **primary storage** and **secondary storage**.
- Primary storage is also known as the main memory and is the memory directly accessible by the CPU.
- Secondary or **external storage is not directly accessible by the CPU.**
- The data from secondary storage **needs to be brought into the primary storage before the CPU can use it.**
- Secondary storage contains a large amount of data permanently.

Arithmetic Logic Unit

- All the **calculations related to the computer system** are performed by the arithmetic logic unit.
- It can perform operations like **addition, subtraction, multiplication, division** etc.
- The **control unit transfers data from storage unit to arithmetic logic unit when calculations need to be performed.**
- The arithmetic logic unit and the control unit **together form the central processing unit.**

Control Unit etc.

- This unit **controls all the other units** of the computer system and so is known as its **central nervous system.**
- It transfers data throughout the computer as required including from storage unit to central processing unit and vice versa.
- The control unit also dictates how the memory, input output devices, arithmetic logic unit etc. should behave.

Computer-System Architecture

Working of the Diagram:

1.Input Phase:

1. Data and instructions are entered into the system through the Input Unit.
2. The Input Unit converts this information into a format that the computer can process and sends it to the Storage Unit for temporary holding.

2.Processing Phase:

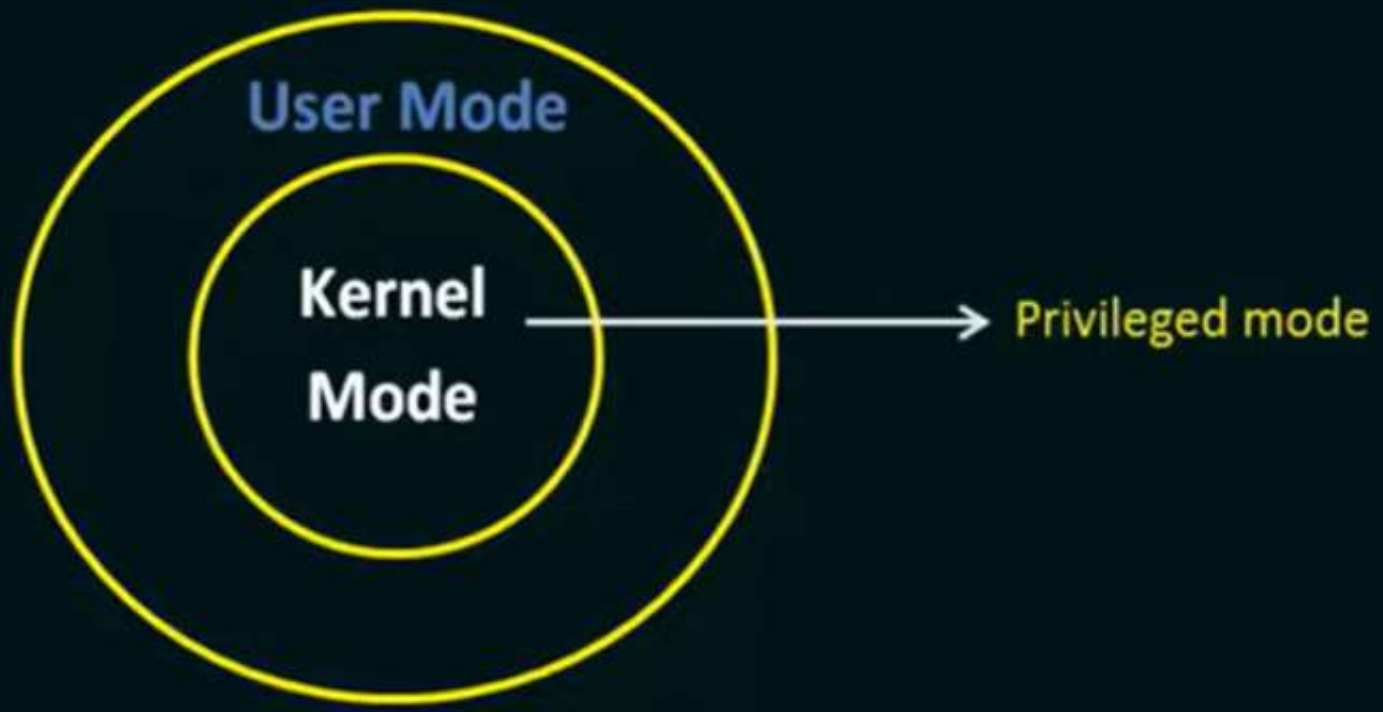
1. The Control Unit fetches instructions from the Storage Unit and interprets them.
2. It then directs the ALU to perform necessary operations on the data.
3. The ALU processes the data and sends the results back to the Storage Unit.

3.Output Phase:

1. Processed data is sent from the Storage Unit to the Output Unit.
2. The Output Unit converts the processed data into a human-

SYSTEM CALL

System calls provide an interface to the services made available by an Operating System.



- System call is the programmatic way in which a computer program requests a service from the kernel of the operating system.
- These calls are generally available as routines written in C and C++.

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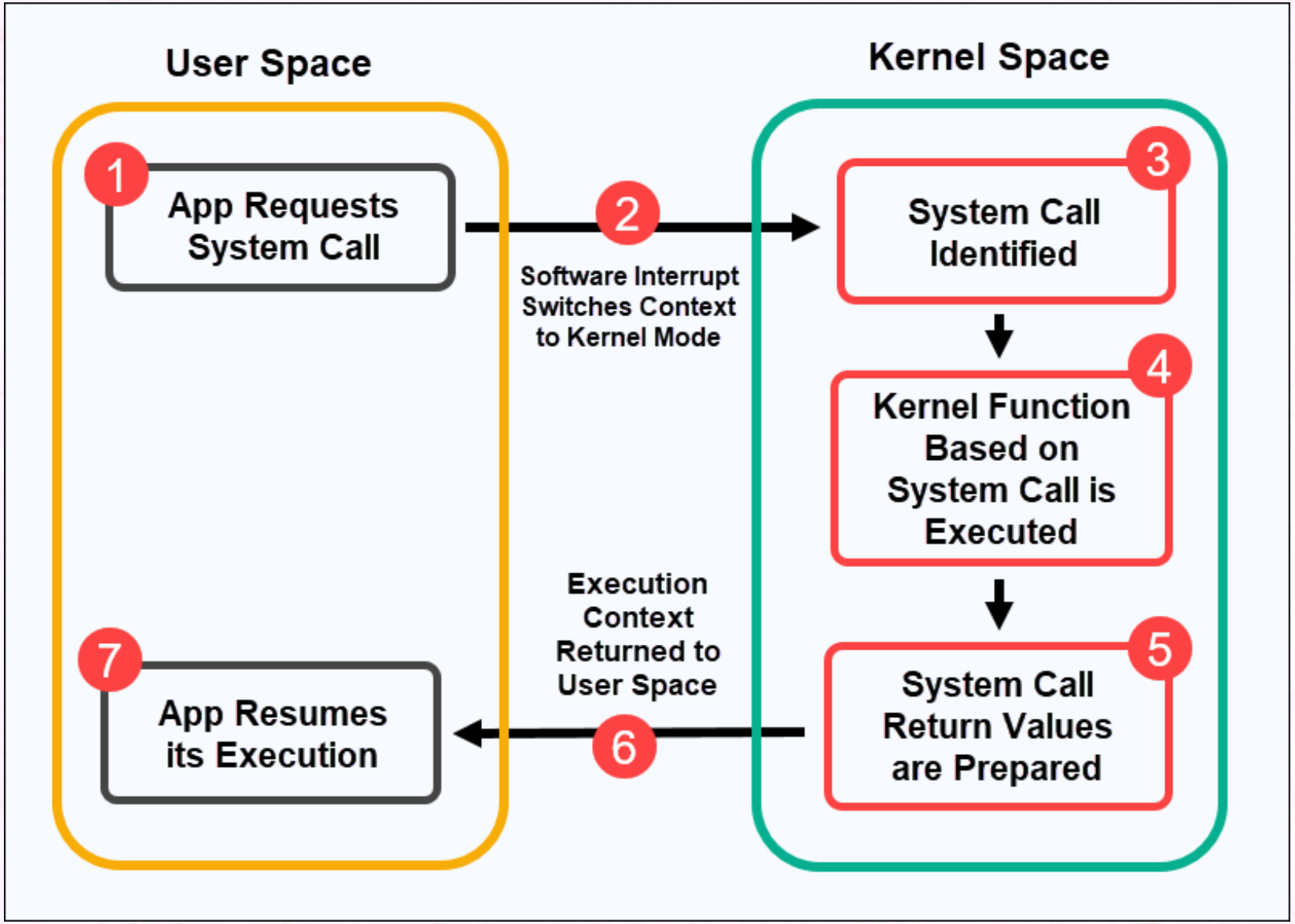
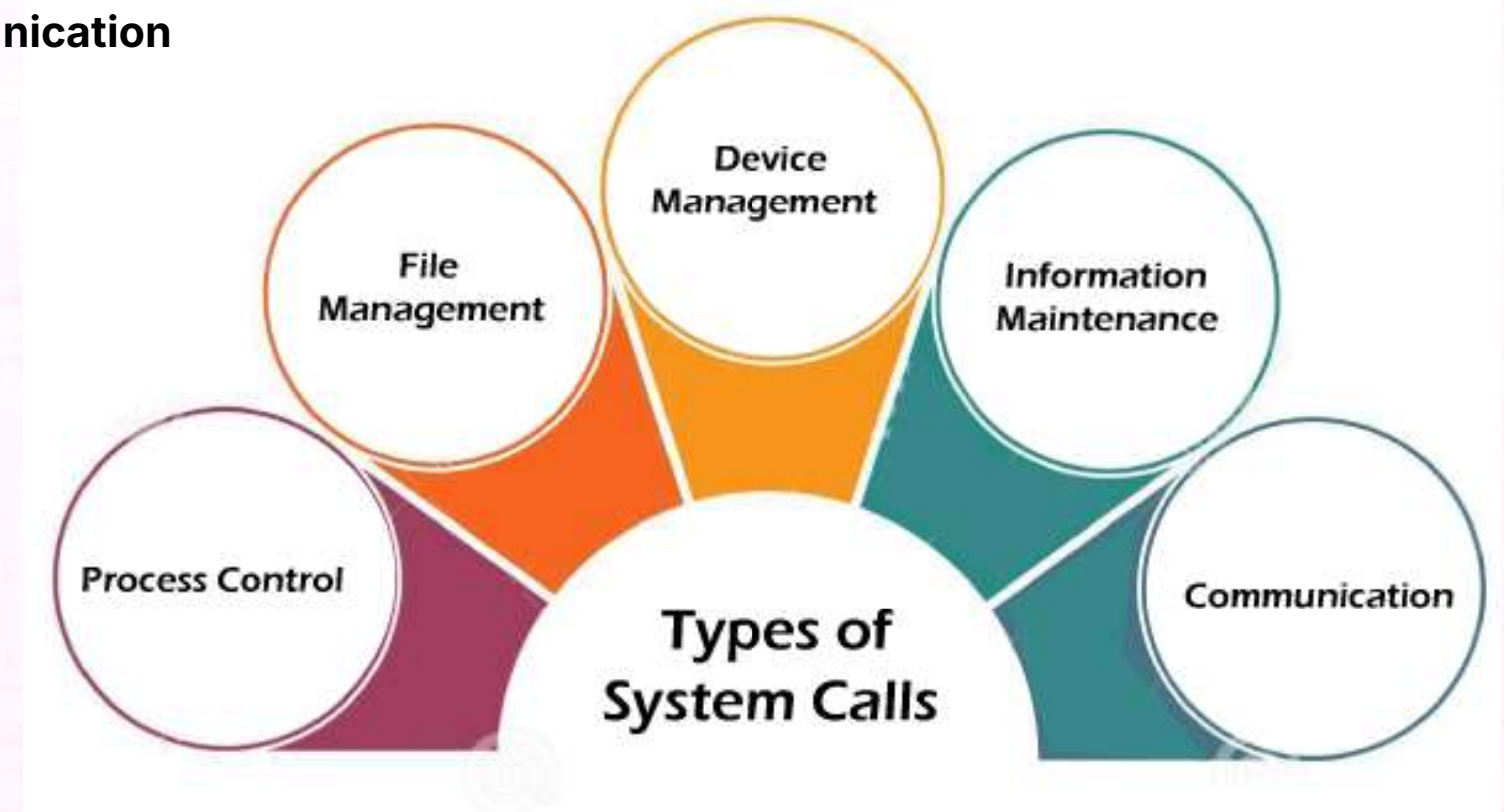

User Space
Kernel Space
1
App Requests System Call
2
Software Interrupt Switches Context to Kernel Mode
3
System Call Identified
4
Kernel Function Based on System Call is Executed
5
System Call Return Values are Prepared
6
Execution Context Returned to User Space
7
App Resumes its Execution

Types of System Calls

1. Process Control
2. File Management
3. Device Management
4. Information Maintenance
5. Communication



These are the example of system calls in os :

Types of SystemCalls	Windows	Linux
	ExitProcess()	exit()
Process Control	CreateProcess()	fork()
	WaitForSingleObject()	wait()
	CreateFile()	open()
File Management	ReadFile()	read()
	WriteFile()	write()
	CloseHandle()	close()
	SetConsoleMode()	ioctl()
Device Management	ReadConsole()	read()
	WriteConsole()	write()
	SetTimer()	alarm()
Information Maintenance	GetCurrentProcessID()	getpid()
	Sleep()	sleep()
	CreatePipe()	pipe()
Communication	CreateFileMapping()	shmget()
	MapViewOfFile()	mmap()

Process Control

Process control is the system call that is used to direct the processes. Some process control examples include creating, load, abort, end, execute, process, terminate the process, etc.

File Management

File management is a system call that is used to handle the files. Some file management examples include creating files, delete files, open, close, read, write, etc.

Device Management

Device management is a system call that is used to deal with devices. Some examples of device management include read, device, write, get device attributes, release device, etc.

Information Maintenance

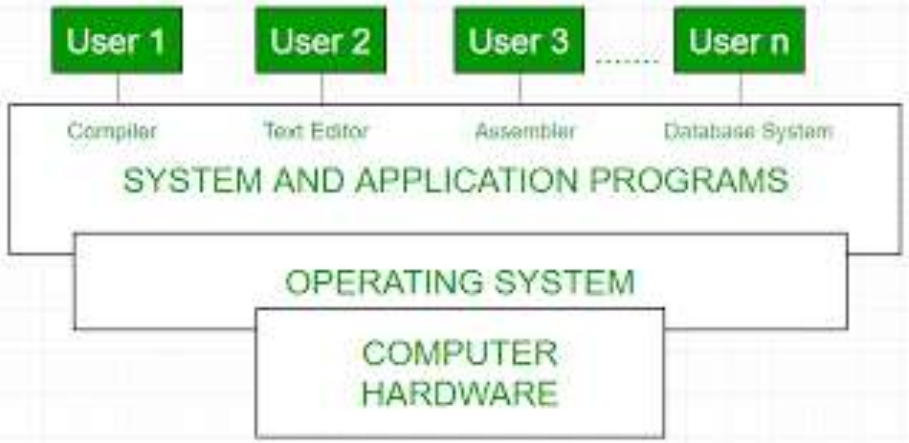
Information maintenance is a system call that is used to maintain information. There are some examples of information maintenance, including getting system data, set time or date, get time or date, set system data, etc.

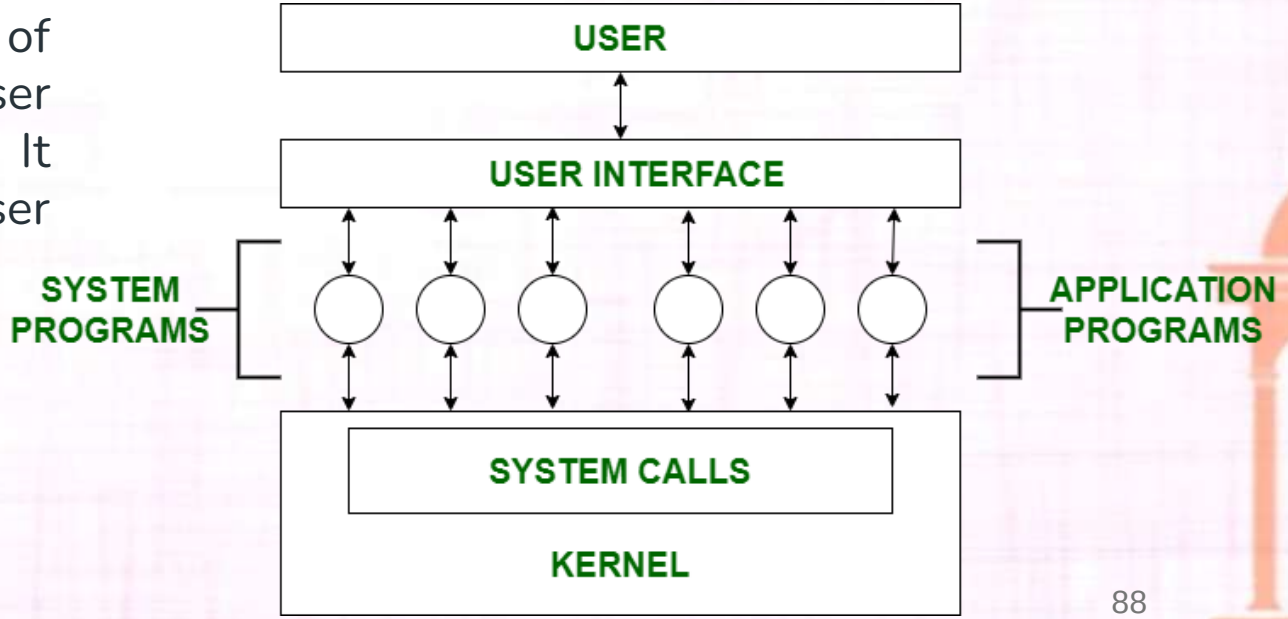
Communication

Communication is a system call that is used for communication. There are some examples of communication, including create, delete communication connections, send, receive messages, etc.

System Programs

System Programming can be defined as the act of building Systems Software using System Programming Languages. According to Computer Hierarchy, one which comes at last is Hardware. Then it is Operating System, System Programs, and finally Application Programs. Program Development and Execution can be done conveniently in System Programs. Some of the System Programs are simply user interfaces, others are complex. It traditionally lies between the user interface and system calls.





SYSTEM PROGRAMS

System programs are software that work with the operating system (OS) to manage a computer's hardware and resources.

System programs communicate and coordinate the activities and functions of hardware and software of a system and also controls the operations of the hardware.

Some examples of system programs are operating system, networking system, web site server, data backup server , device drivers, middleware, utilities, and programming language interpreters.

File Management

- Create
- Delete
- Copy
- Rename
- Print
- Dump
- List, and generally manipulate files and directories.

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Status Information

Ask the system for:

- Date, Time
- Amount of available memory or disk space
- Number of users
- Detailed performance
- Logging, and debugging information etc.

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Programming-language support

- Compilers
- Assemblers
- Debuggers and
- Interpreters

for common programming languages

(such as C, C++, Java, Visual Basic, and PERL)

are often provided to the user with the operating system.

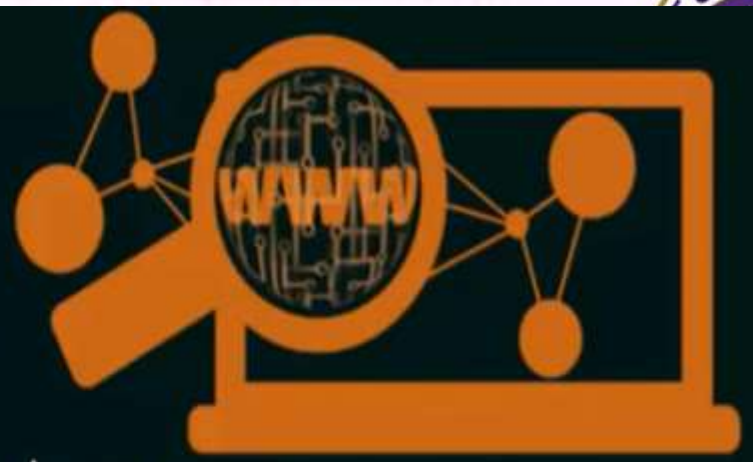
Program loading and execution

Once a program is assembled or compiled, it must be loaded into memory to be executed.

The system may provide:

- Absolute loaders
- Relocatable loaders
- Linkage editors and
- Overlay loaders

Communications



These programs provide the mechanism for:

- Creating virtual connections among processes, users, and computer systems.
- Allowing users to send messages to one another's screens
- To browse webpages
- To send electronic-mail messages
- To log in remotely or to transfer files from one machine to another.

Examples:

Web Browsers

Word Processors

Spreadsheets

Database Systems

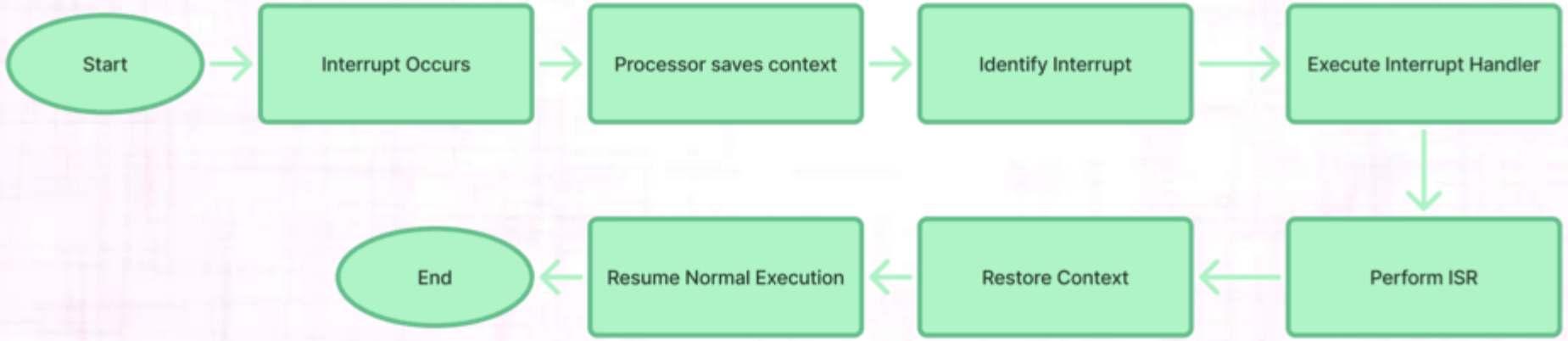
Games

etc.

Application Programs

An **interrupt** is a signal emitted by a device attached to a computer or from a program within the computer. It requires the operating system (OS) to stop and figure out what to do next. An interrupt temporarily stops or terminates a service or a current process.

Flowchart



- Step 1:- Any time that an interrupt is raised, it may either be an I/O interrupt or a system interrupt.

- Step 2:- The current state comprising registers and the program counter is then stored in order to conserve the state of the process.

- Step 3:- The current interrupt and its handler is identified through the interrupt vector table in the processor.

- Step 4:- This control now shifts to the interrupt handler, which is a function located in the kernel space.

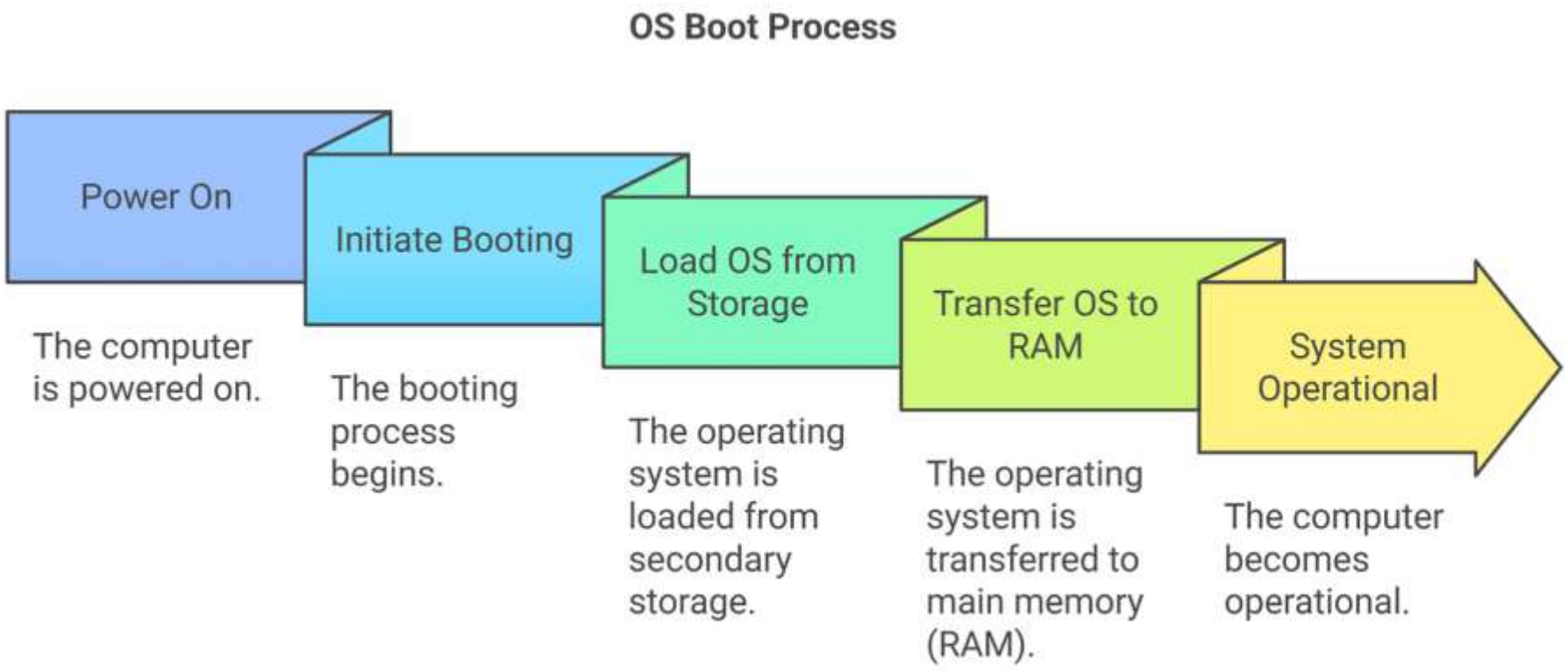
- Step 5:- Specific tasks are performed by Interrupt Service Routine (ISR) which are essential to manage interrupt.

- Step 6:- The status from the previous session is retrieved so as to build on the process from that point.

- Step 7:- The control is then shifted back to the other process that was pending and the normal process continues.

System Boot

Booting is the process that occurs when a computer is powered on. During this process, a mechanism within the system loads the operating system from secondary storage into the main memory (RAM), enabling the computer to become operational.



System Boot

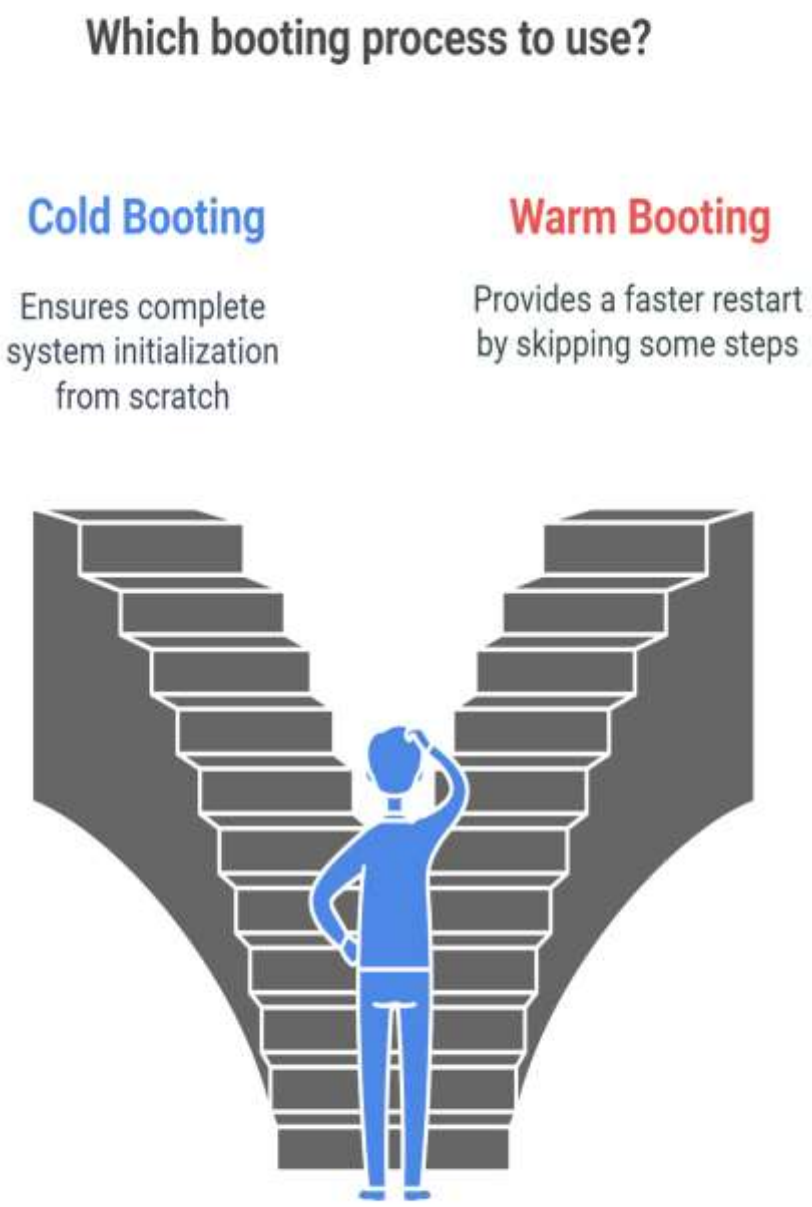
Cold or Hard Booting

Cold booting, also known as a hard boot, refers to the process of powering on a computer that is completely switched off. During this procedure, the system performs a full power-on self-test (POST) to initialize hardware devices and loads the operating system from secondary storage into random-access memory (RAM). This ensures a complete system startup from scratch.

Soft or Warm Booting

Warm booting, also called a soft boot or restart, involves rebooting a computer without fully powering it down. This process is typically initiated through an operating system restart command or a specific key combination. Unlike cold booting, warm booting skips some hardware initialization steps, as the hardware components remain powered and were previously initialized.

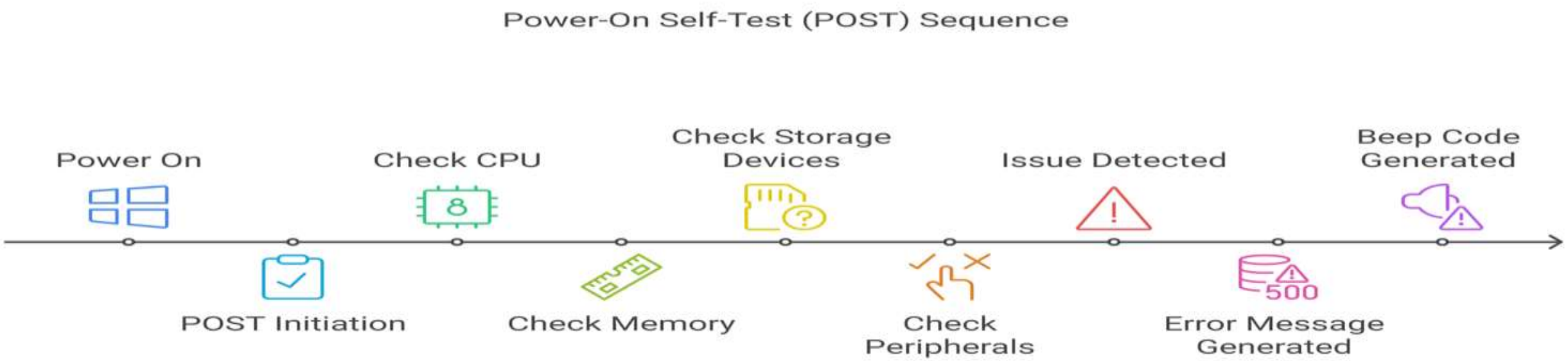
Both cold and warm booting are essential for computer operation. Cold booting ensures a complete system initialization, while warm booting provides a faster restart option by bypassing the full startup sequence.



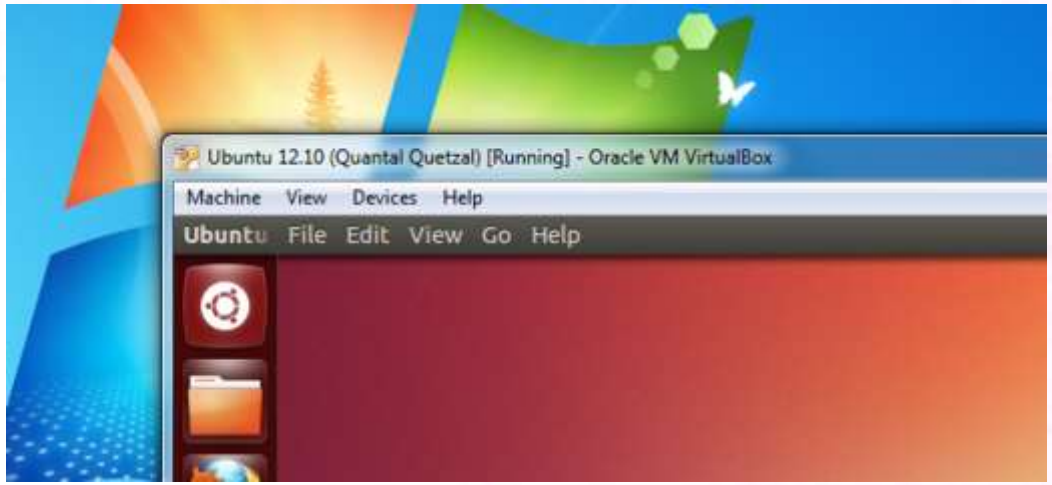
System Boot

What is Power-On Self-Test (POST) Booting?

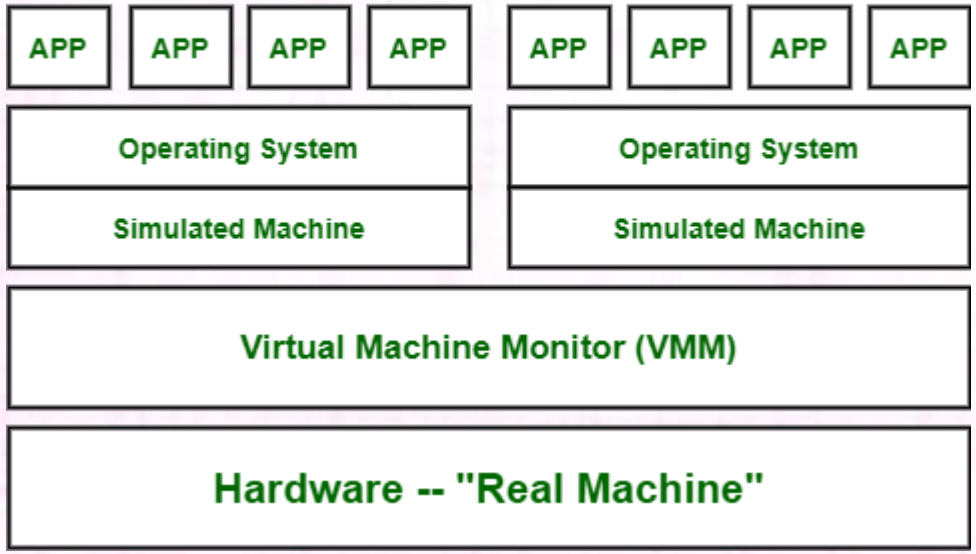
The Power-On Self-Test (POST) is the initial diagnostic process performed during a computer's booting cycle. Each time the system is powered on, POST checks the presence and functionality of critical hardware components, such as the CPU, memory, storage devices, and peripherals. This routine ensures that the hardware is operational and ready for use. If any issues are detected, POST generates error messages or beep codes to indicate the problem. Once POST completes successfully, the system proceeds to load the operating system and necessary software for normal operation.



VIRTUAL MACHINE



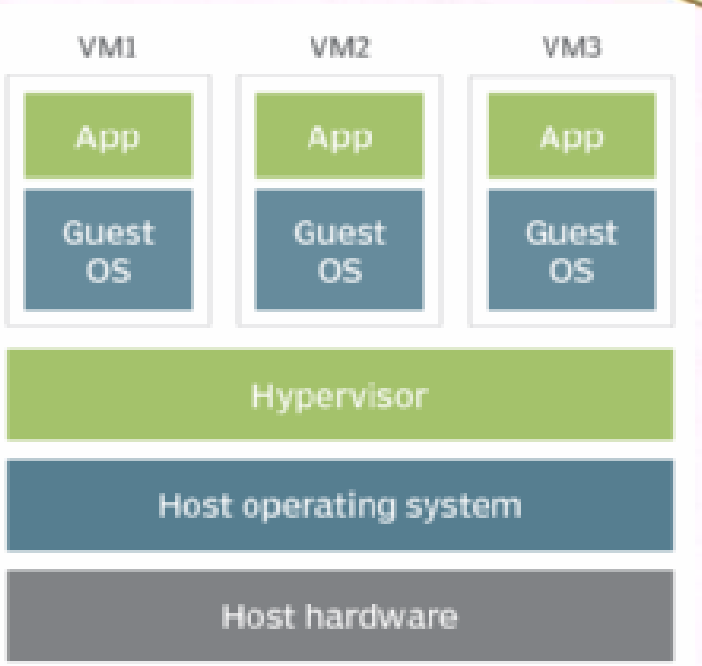
System Virtual Machine



VIRTUAL MACHINE

Virtual Machine can be defined as an emulation of the computer systems in computing.

Virtual Machine is based on computer architectures.

It also gives the functionality of physical computers.

The implementation of VM may consider ...
specialized software, hardware, or a combination of both.